EMCH 574 – Vibration and Waves
Homework 02

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Due date TBD

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Assignment

- G signifies students that take this class for Graduate Credit; they must do all the items.
- UG students are not required to do the G work, but they may do it for bonus points.
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional; they will earn bonus points.

Notes

- Show your work to obtain partial credit if appropriate.
- Always display input data!
- Partial credit might be given based on work done and MATLAB codes if they are attached; without them, no partial credit can be given.
- Display numerical results to three significant digits (four if the first digit is 1).
- Use SI system of measurement reference.
- Scale units using appropriate prefixes (e.g., nano, micro, milli, kilo, mega, giga, and so on) reference.
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional. They may be solved for extra credit.
- Normalized plots are plots with the y-axis scaled between -1 and $+1$. A plotted variable is normalized by dividing it by its maximum absolute value.

Notations and Abbreviations

- BVP = boundary value problem.
- EOM = equation of motion.
- FBD = free body diagram.
- FRF = frequency response function.
- IVP = initial value problem.
- LHS = left hand side.
- NLM = Newton's law of motion.
- ODE = ordinary differential equation.
- PDE = partial differential equation.
- RHS = right hand side.

1 Waves

A.1 Wave Fundamentals

(1) List at least five examples of waves met in everyday life.

(2) Explain in your own words the difference between disturbance waves and continuous waves.

(3) Explain using text and sketches the definition of wavespeed for a disturbance wave.

(4) Give at least three different equivalent expressions for a disturbance wave.

(5) A car pileup takes place on an interstate and propagates as a wave for several miles. What type of wave is this? Choose one of the following:

- **(a)** Continuous wave.
- **(b)** Disturbance wave.

(6) Explain using text and sketches the definition of wavespeed for a continuous harmonic wave.

(7) Explain using text and sketches the definition of wavelength and wavenumber for a continuous harmonic wave.

(8) Give at least three different equivalent expressions for a continuous harmonic wave.

(9) What do waves carry? Choose one of the following:

- **(a)** Matter.
- **(b)** Energy.
- **(c)** Both.
- **(d)** None of the above.

(10) Graduate work. Write about the particle-wave duality of light.

Solution to A.1

1. Everyday examples of waves. Sound traveling through air, ripples spreading across water, light from a lamp, a pulse moving along a rope, and seismic waves traveling through Earth are familiar examples. Radio and Wi-Fi signals are also electromagnetic waves.

$$\boxed{\begin{array}{c}\text{Sound, water ripples, light, rope pulses,}\\ \text{seismic waves, and radio signals.}\end{array}} \tag{A.1.1}$$

2. Disturbance waves and continuous waves. A disturbance wave begins with a localized change, such as a single pulse, and carries that profile through a medium. A continuous wave is produced when the source oscillates repeatedly, creating a sustained sequence of crests and troughs. The distinction concerns how the source moves over time; both types propagate through space.

$$\boxed{\begin{array}{c}\text{A disturbance is localized in time.}\\ \text{A continuous wave is sustained by repeated oscillation.}\end{array}} \tag{A.1.2}$$

3. Wavespeed of a disturbance. Wavespeed is the distance a recognizable point on the disturbance travels divided by the elapsed time. For the two profiles below, the pulse advances by Δx during Δt, so its speed is $c = \Delta x/\Delta t$. For a right-moving pulse, $u(x,t) = F(x-ct)$ keeps the same shape because a fixed feature satisfies $x - ct = \text{constant}$.

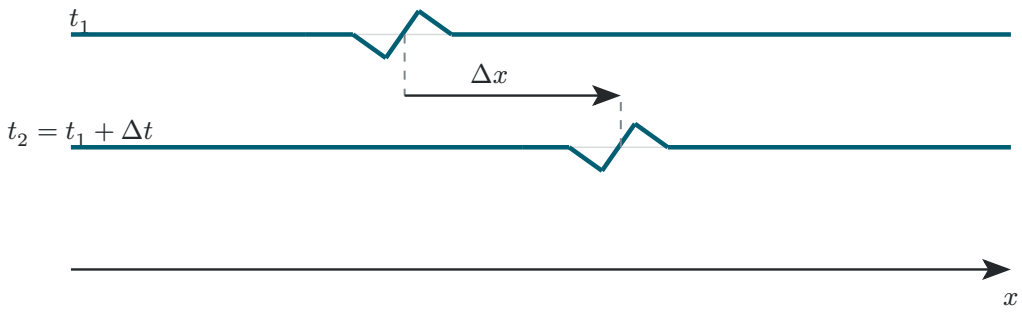


Figure 1: *A localized disturbance advances by Δx during Δt.*

$$\boxed{c = \frac{\Delta x}{\Delta t}} \tag{A.1.3}$$

4. Equivalent disturbance-wave expressions. For a right-traveling disturbance in a uniform medium with wavespeed c, let F describe its profile and let $\ell > 0$ be a reference length. Defining $G(\tau) = F(-c\tau)$ and $H(\xi) = F(\ell\xi)$ gives equivalent profiles written with a spatial, temporal, or dimensionless argument. Each expression therefore describes the same shape translating at speed c.

$$\boxed{u(x,t) = F(x-ct) = G\left(t - \frac{x}{c}\right) = H\left(\frac{x-ct}{\ell}\right)} \tag{A.1.4}$$

5. Interstate pileup. A car pileup begins locally and spreads as vehicles encounter the slowdown. It is therefore a disturbance wave.

$$\boxed{\text{Disturbance wave (choice b).}} \qquad \text{(A.1.5)}$$

6. Wavespeed of a continuous harmonic wave. A harmonic wave has a sinusoidal profile that repeats as it propagates. Wavespeed is the speed of a fixed phase point, such as a crest. If that crest moves by Δx during Δt, then $c = \Delta x/\Delta t$. For $u(x,t) = A\cos(\gamma x - \omega t + \phi)$, constant phase gives $\gamma x - \omega t + \phi = \text{constant}$ and hence $c = \omega/\gamma$.

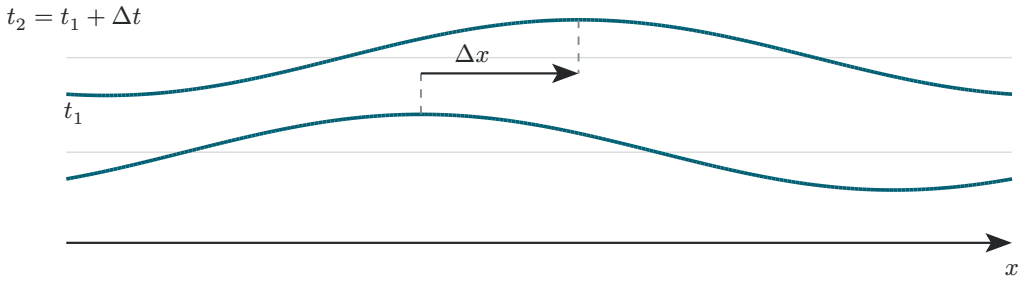


Figure 2: *A fixed phase point advances with the continuous harmonic profile.*

$$\boxed{c = \frac{\Delta x}{\Delta t} = \frac{\omega}{\gamma} = f\lambda} \qquad \text{(A.1.6)}$$

7. Wavelength and wavenumber. Wavelength λ is the spatial distance over which the harmonic profile completes one cycle, measured between adjacent points in the same phase, such as neighboring crests. Wavenumber γ is the phase change per unit distance. Since one spatial cycle advances the phase by 2π radians over distance λ, the relationship is $\gamma = 2\pi/\lambda$. Its SI unit is radians per meter, commonly written rad/m.

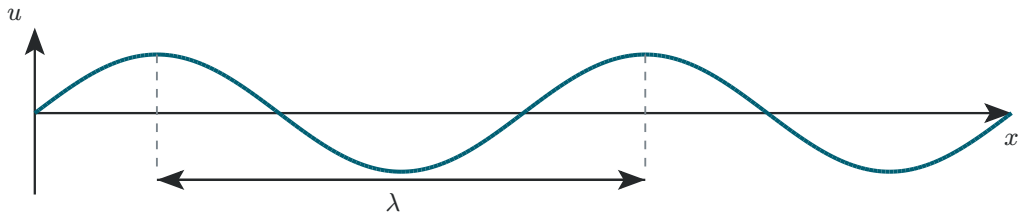


Figure 3: *The wavelength spans one complete spatial cycle between matching phase points.*

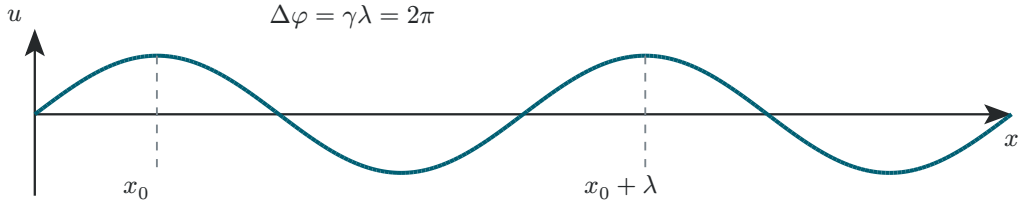


Figure 4: *Across one wavelength, the harmonic phase increases by 2π radians.*

$$\boxed{\gamma = \frac{2\pi}{\lambda}, \qquad \lambda = \frac{2\pi}{\gamma}} \tag{A.1.7}$$

8. Equivalent continuous harmonic-wave expressions. A traveling harmonic wave can be written as a cosine with phase, a sine with an adjusted phase, or as the real part of a complex exponential. These forms describe the same oscillatory motion with suitable choices of amplitude and phase. Here A is the amplitude, ω is angular frequency, γ is wavenumber, and ϕ is phase.

$$\boxed{\begin{aligned} u(x,t) &= A\cos(\gamma x - \omega t + \phi) \\ &= A\sin(\gamma x - \omega t + \phi + \tfrac{\pi}{2}) \\ &= \mathrm{Re}\Big\{ A e^{i(\gamma x - \omega t + \phi)} \Big\} \end{aligned}} \tag{A.1.8}$$

9. What waves carry. A wave transports energy through space or through a medium. In a mechanical wave, particles of the medium oscillate around their equilibrium positions; the wave does not carry the medium along as a net flow of matter.

$$\boxed{\text{Energy (choice b).}} \tag{A.1.9}$$

10. Graduate work. Particle-wave duality of light. Light propagates and interferes as an electromagnetic wave, yet exchanges energy with matter in discrete photons. A photon of frequency f carries energy $E = hf$, where h is Planck's constant; for light in vacuum its momentum is $p = h/\lambda$ [4]. The photoelectric effect shows that energy transfer occurs in photon-sized packets, while interference and diffraction reveal wave behavior. These observations reveal complementary aspects of light. Its wave behavior governs propagation and interference, while photon events describe localized exchanges with detectors and matter [3].

$$\boxed{E = hf, \qquad p = \frac{h}{\lambda}} \tag{A.1.10}$$

A.2 Simple Wave Calculations

Use the electromagnetic wavespeed $c_{\text{light}} = 300{,}000\,\text{km/s}$ and sound wavespeed $c_{\text{sound}} = 340\,\text{m/s}$.

Do the following.

(1) Calculate the frequency of visible light with typical wavelength $\lambda = 510\,\text{nm}$.

(2) Calculate the Hz frequency f of an AM radio wave with wavelength $\lambda = 550\,\text{m}$.

(3) Infrared radiation has a typical wavelength $\lambda = 10\,\mu\text{m}$. What is the typical frequency of infrared radiation?

(4) Calculate the wavelength λ of an FM radio wave with frequency $f = 89\,\text{MHz}$.

(5) The lightning flash is observed at 8:23:32 p.m., whereas the thunder is heard at 8:23:36 p.m. How far away is the lightning strike? Give your answer in kilometers and miles.

(6) One of Verizon's 5G mobile signals has frequency $f \approx 29\,\text{GHz}$. What is the corresponding wavelength?

(7) A microwave oven typically operates at $f = 2.5\,\text{GHz}$.

- **(a)** What is the wavelength λ?
- **(b)** What is the maximum permissible hole size D in the oven's door mesh, given that the largest hole in a Faraday cage must be at most $\lambda/2$?
- **(c)** Measure approximately the hole size in a microwave oven door mesh and discuss your results. You may search the web for various opinions on this matter.

(8) Graduate work. Given that the diameter of an electron is estimated to be $D = 2\,\text{pm}$, complete items (a) and (b).

- **(a)** Calculate the frequency of the electromagnetic wave that has a wavelength equal to the electron diameter.
- **(b)** In what spectrum range would this electromagnetic wave be? (options: infrared, visible, ultraviolet, X-ray, gamma ray).

Solution to A.2

In one period T, a wave crest travels one wavelength λ. Since $f = 1/T$, the wavespeed relation is

$$c = \frac{\lambda}{T} = \lambda f, \qquad f = \frac{c}{\lambda}, \qquad \lambda = \frac{c}{f}.$$

The supplied speeds in SI units are $c_{\text{light}} = 3.00 \times 10^8$ m/s and $c_{\text{sound}} = 340$ m/s. These values are used throughout; unit prefixes are converted before substitution.

For A.2.1, the visible-light wavelength is $\lambda = 510$ nm $= 510 \times 10^{-9}$ m. Therefore,

$$\boxed{f = \frac{3.00 \times 10^8}{510 \times 10^{-9}} \text{ Hz} = 5.88 \times 10^{14} \text{ Hz} = 588 \text{ THz}} \qquad \text{(A.2.1)}$$

For A.2.2, the AM wavelength is $\lambda = 550$ m. Using the same relation gives

$$\boxed{f = \frac{3.00 \times 10^8}{550} \text{ Hz} = 5.45 \times 10^5 \text{ Hz} = 545 \text{ kHz}} \qquad \text{(A.2.2)}$$

For A.2.3, the infrared wavelength is $\lambda = 10\ \mu$m $= 10 \times 10^{-6}$ m. Its frequency is

$$\boxed{f = \frac{3.00 \times 10^8}{10 \times 10^{-6}} \text{ Hz} = 3.00 \times 10^{13} \text{ Hz} = 30.0 \text{ THz}} \qquad \text{(A.2.3)}$$

The next item supplies frequency instead of wavelength. For A.2.4, $f = 89$ MHz $= 89 \times 10^6$ Hz, so

$$\boxed{\lambda = \frac{3.00 \times 10^8}{89 \times 10^6} \text{ m} = 3.37 \text{ m}} \qquad \text{(A.2.4)}$$

For A.2.5, the observed delay between the flash and thunder is $\Delta t = 36 - 32 = 4$ s. Taking the light and sound to originate at the same instant and travel the same distance d, their arrival-time difference is

$$\Delta t = \frac{d}{c_{\text{sound}}} - \frac{d}{c_{\text{light}}}.$$

Solving this equation for distance accounts for both travel times and gives

$$d = \frac{\Delta t}{1/c_{\text{sound}} - 1/c_{\text{light}}} = \frac{4}{1/340 - 1/(3.00 \times 10^8)} \text{ m}.$$

Using 1 km $= 1000$ m and the exact conversion 1 mi $= 1609.344$ m, the required results are

$$\boxed{d = 1.360 \text{ km} = 0.845 \text{ mi}} \qquad \text{(A.2.5)}$$

Because light travels much faster than sound, the usual approximation $d \simeq c_{\text{sound}}\Delta t = 340(4) = 1360$ m gives the same reported answers. The finite-light correction is only 1.541 mm.

For A.2.6, the supplied mobile-signal frequency is $f \approx 29$ GHz $= 29 \times 10^9$ Hz. Its wavelength is

$$\boxed{\lambda = \frac{3.00 \times 10^8}{29 \times 10^9}\ \text{m} = 10.34\ \text{mm}} \qquad \text{(A.2.6)}$$

The microwave oven in A.2.7 operates at $f = 2.5$ GHz $= 2.5 \times 10^9$ Hz. Applying the same wavespeed relation answers part (a),

$$\boxed{\lambda = \frac{3.00 \times 10^8}{2.5 \times 10^9}\ \text{m} = 0.1200\ \text{m} = 120.0\ \text{mm}} \qquad \text{(A.2.7.a)}$$

For part (b), the problem specifies $D \leq \lambda/2$. Consequently, the largest diameter allowed by that stated criterion is

$$\boxed{D_{\text{max}} = \frac{\lambda}{2} = \frac{0.1200}{2}\ \text{m} = 60.0\ \text{mm}} \qquad \text{(A.2.7.b)}$$

For A.2.7.c, we measured the mesh on our Oster small office microwave and found a hole diameter of roughly 1.2 mm and a center-to-center pitch of roughly 1.6 mm. Using the measured diameter gives

$$\boxed{D_{\text{mesh}} \approx 1.2\ \text{mm}, \qquad \frac{D_{\text{mesh}}}{\lambda} \approx 0.01000, \qquad \frac{D_{\text{max}}}{D_{\text{mesh}}} \approx 50.0} \qquad \text{(A.2.7.c)}$$

Thus, the measured hole is approximately one hundredth of the microwave wavelength and fifty times smaller than the classroom upper bound. Its opening is nevertheless much larger than a visible-light wavelength, so food remains visible through the perforated screen. The 60.0 mm result answers the supplied $\lambda/2$ criterion; it is not a microwave-door design specification.

For the graduate calculation in A.2.8.a, use the stipulated length $D = 2$ pm $= 2 \times 10^{-12}$ m as the electromagnetic wavelength. Then

$$\boxed{f = \frac{c_{\text{light}}}{D} = \frac{3.00 \times 10^8}{2 \times 10^{-12}}\ \text{Hz} = 1.500 \times 10^{20}\ \text{Hz} = 150.0\ \text{EHz}} \qquad \text{(A.2.8.a)}$$

This calculation uses the supplied length scale without establishing a physical electron diameter. The resulting wavelength is 0.00200 nm, below the 0.01 nm gamma-ray range described by NASA[1]. Therefore, the wavelength-based classification requested in A.2.8.b is

$$\boxed{\text{Gamma ray}} \qquad \text{(A.2.8.b)}$$

MATLAB code. *Script used to reproduce the numerical results in A.2.*

```
% A.2. Frequencies, wavelengths, and lightning distance.
clear; clc;
c = 3e8; cs = 340;                % light and sound speeds, m/s
lv = 510e-9; la = 550; li = 10e-6; % wavelengths, m
ff = 89e6; fg = 29e9; fm = 2.5e9; % frequencies, Hz
dt = 36-32;                       % lightning arrival delay, s
D = 1.2e-3; p = 1.6e-3;           % measured mesh diameter and pitch, m
de = 2e-12;                       % supplied electron length, m
disp(table(c,cs,lv,la,li,ff,fg,fm,dt,D,p,de));

% A.2.1--4. Use c = lambda*f.
fv = c/lv; fa = c/la; fi = c/li; lf = c/ff;
fprintf('Visible frequency = %.6g Hz\n',fv);
fprintf('AM frequency = %.6g Hz\n',fa);
fprintf('Infrared frequency = %.6g Hz\n',fi);
fprintf('FM wavelength = %.6g m\n',lf);

% A.2.5. Sound travel time minus light travel time is dt.
d = dt/(1/cs-1/c);
fprintf('Lightning distance = %.6g km = %.6g mi\n', ...
    d/1000,d/1609.344);

% A.2.6--7. Mobile and microwave wavelengths.
lg = c/fg; lm = c/fm; dm = lm/2;
fprintf('5G wavelength = %.6g mm\n',1000*lg);
fprintf('Microwave wavelength = %.6g mm\n',1000*lm);
fprintf('Maximum hole diameter = %.6g mm\n',1000*dm);
fprintf('Measured Oster hole diameter = %.3g mm; pitch = %.3g mm\n', ...
    1000*D,1000*p);
fprintf('Hole/wavelength = %.6g; limit/hole = %.6g\n',D/lm,dm/D);

% A.2.8. Use the supplied length as the wavelength.
fe = c/de;
fprintf('Electron-length frequency = %.6g Hz (gamma ray)\n',fe);
```

A.3 Calculate Car Motion on a Wavy Road

Consider a car traveling on a wavy road. The road surface undulates with an amplitude $a = 2.5\,\text{in}$ and a wavelength $\lambda = 105\,\text{ft}$ that follows a cosine shape. The car is modeled as a spring-mass-damper (Figure 5) with natural frequency $f_n = 0.95\,\text{Hz}$ and damping ratio $\zeta = 22\%$. The maximum car speed is $v_{\text{max}} = 125\,\text{mph}$.

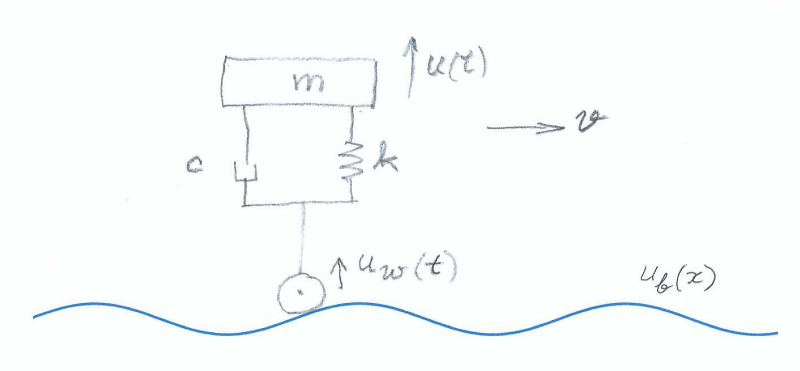


Figure 5: *Car riding on a wavy road.*

Do the following.

(1) Display input data.

(2) Calculate wavy road wavenumber γ.

(3) Plot the wavy road profile $u_b(x)$ for $N_\lambda = 2$ wavelengths.

(4) Calculate wavy-road frequency f_{max} for a maximum car speed of v_{max} and define the frequency range $f = \text{linspace}(0, f_{\text{max}}, N_f)$ with $N_f = 2000$.

(5) Plot $|\text{FRF}(f; f_n)|$. Use datatips to locate the peak response and the matched-frequency response.

(6) Calculate the matched-frequency response value $|\text{FRF}_n|$. Compare with the matched-frequency readings at item (5).

(7) Calculate the car speed v_n needed to make wavy-road frequency f match the car natural frequency f_n.

(8) Use the conversion $f = v/\lambda$ to plot $|\text{FRF}(v; f_n)|$. Use datatips on the plot to find:

- **(a)** Find v_n and $|\text{FRF}_n|$, and compare them with the results of items (6) and (7).
- **(b)** Find v_{cr} and $|\text{FRF}_{\text{max}}|$.
- **(c)** The limit speed v_{limit} for which the motion amplification is limited to $|\text{FRF}_{\text{lim}}| = 1.5$.

(9) Graduate work. Find the road wavelength λ_{new} that will make $v_n = 35\,\text{mph}$.

(10) Graduate work. Repeat items (5) through (8) for a car with worn-out shocks that provide only $\zeta = 10\%$, and compare the two cases.

Solution to A.3

A.3.1. Input data. I measure the body coordinate $u(t)$ and road coordinate $u_b(t)$ upward from static equilibrium. The spring and damper act through their relative deformation, so the wheel displacement is the excitation input. The supplied diagram therefore represents base displacement excitation rather than an applied force.

Quantity	Value	Unit
a	0.0635	m
λ	32.004	m
f_n	0.950	Hz
ζ	0.220	1
$v_{\max}$	55.88	m/s
N_λ	2	wavelengths
N_f	2000	points
$\lvert H\rvert_{\lim}$	1.50	1

For the equations of motion, the spring and damper forces depend on $u - u_b$ and $\dot{u} - \dot{u}_b$. Newton's law consequently gives

$$m\ddot{u} + c(\dot{u} - \dot{u}_b) + k(u - u_b) = 0, \qquad m\ddot{u} + c\dot{u} + ku = c\dot{u}_b + ku_b.$$

With $\omega_n = \sqrt{k/m}$ and $c/m = 2\zeta\omega_n$, harmonic inputs $u_b = \Re\{\hat{U}_b e^{i\omega t}\}$ give the absolute body-to-road displacement ratio

$$H(r) = \frac{\hat{U}}{\hat{U}_b} = \frac{1 + i2\zeta r}{1 - r^2 + i2\zeta r}, \qquad r = \frac{f}{f_n}.$$

Its magnitude is

$$|H(r)| = \sqrt{\frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2}}.$$

The numerator follows from the base-motion terms on the right side of the equation of motion. It is also the absolute displacement transmissibility derived in the course notes (pp. 56–59).

A.3.2. Wavy-road wavenumber. One spatial period advances the phase by 2π, so

$$\gamma = \frac{2\pi}{\lambda} = \frac{2\pi}{32.004\ \text{m}} = 0.196325\ \text{rad/m}.$$

$$\boxed{\gamma = 0.1963\ \text{rad/m}} \qquad \text{(A.3.2)}$$

A.3.3. Road profile. The supplied cosine starts at a crest, hence $u_b(x) = a\cos(\gamma x)$. The following figure spans the requested two wavelengths, or 64.008 m.

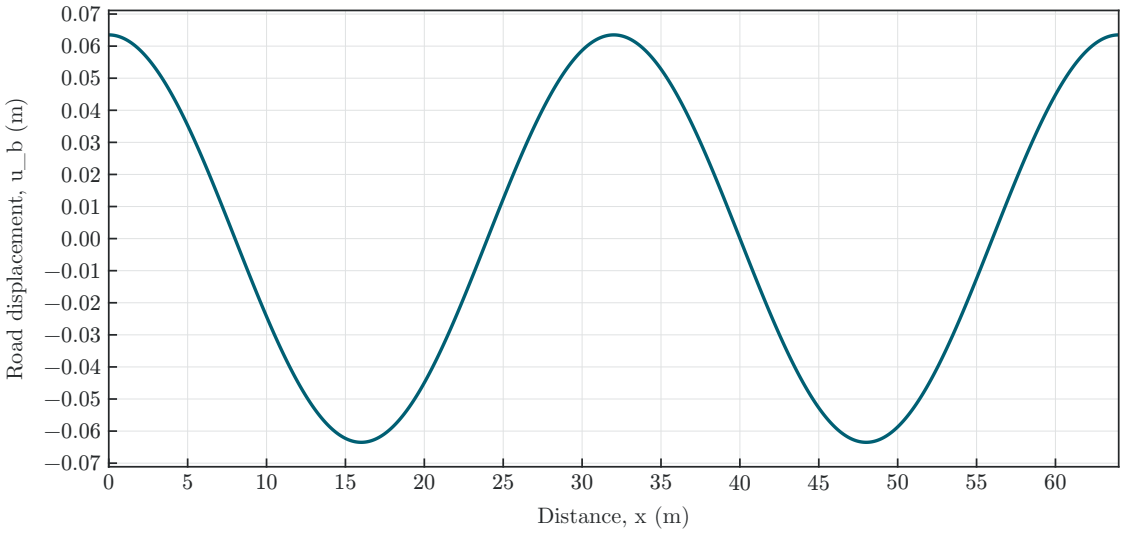


Figure 6: *A.3.3. Cosine road profile across two wavelengths.*

$$\boxed{u_b(x) = 0.0635 \cos\left(\frac{2\pi x}{32.004 \text{ m}}\right) \text{ m}, \quad 0 \le x \le 64.008 \text{ m}} \qquad \text{(A.3.3)}$$

A.3.4. Road-frequency range. At constant speed, $x = vt$ makes the road input $u_b(t) = a\cos(2\pi vt/\lambda)$, so $f = v/\lambda$. The maximum road frequency is therefore

$$f_{\max} = \frac{v_{\max}}{\lambda} = \frac{55.88 \text{ m/s}}{32.004 \text{ m}} = 1.74603 \text{ Hz}.$$

I evaluate $f = \text{linspace}(0, 1.74603 \text{ Hz}, 2000)$ for the plotted curves.

$$\boxed{f_{\max} = 1.746 \text{ Hz}, \qquad N_f = 2000} \qquad \text{(A.3.4)}$$

A.3.5. Healthy-shock frequency response. The figure plots the absolute body displacement transmissibility for healthy shocks. The matched-frequency marker is evaluated at exactly $f = f_n$, while the peak marker is evaluated at the analytic stationary point; neither marker is restricted to the sampled curve grid.

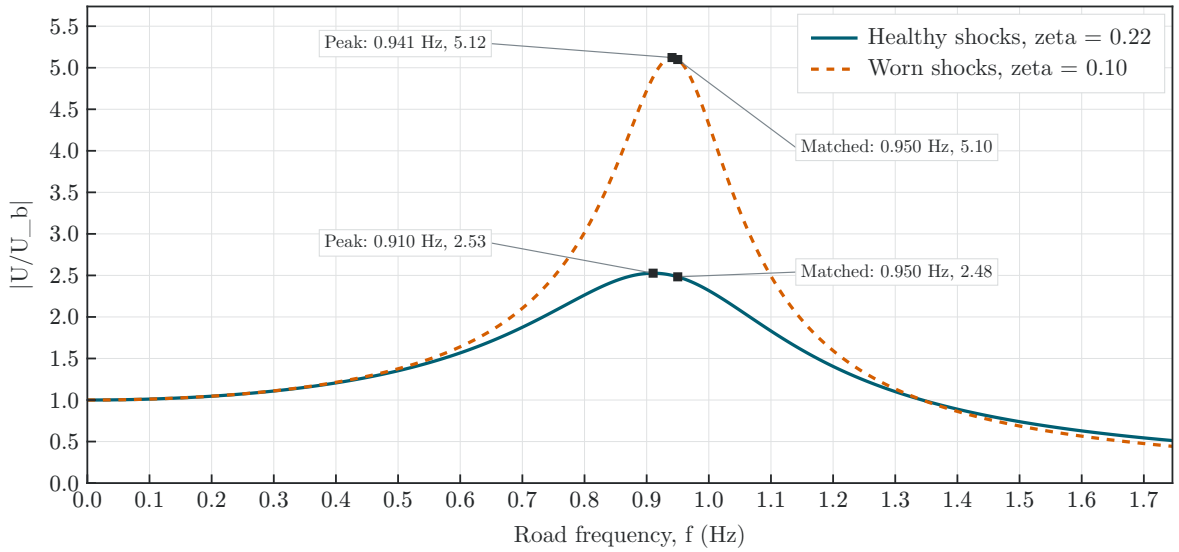


Figure 7: *A.3.5 and A.3.10. Absolute body-to-road displacement FRF versus road frequency. The curves compare healthy and worn shocks; square markers show analytic peak and matched-frequency events.*

A.3.6. Matched-frequency response. At $r = 1$, the complex response is $H(1) = 1 - i/(2\zeta)$, and its magnitude is

$$|H_n| = \sqrt{1 + \frac{1}{(2\zeta)^2}} = \sqrt{1 + \frac{1}{(0.440)^2}} = 2.48300.$$

The plotted analytic marker agrees with this value; linear interpolation on the 2000-point frequency curve gives the same value to the reported precision. For comparison, the true healthy maximum is 2.52688 at $f = 0.910396$ Hz, below f_n.

$$\boxed{\begin{aligned} |H_n| &= 2.48 \quad \text{at } f = f_n = 0.950 \text{ Hz}, \\ |H|_{\max} &= 2.53 \quad \text{at } f = 0.910 \text{ Hz}. \end{aligned}} \tag{A.3.6}$$

A.3.7. Matched-frequency speed. Since $f = v/\lambda$, setting $f = f_n$ gives

$$v_n = f_n \lambda = (0.950 \text{ Hz})(32.004 \text{ m}) = 30.4038 \text{ m/s} = 68.0114 \text{ mph}.$$

$$\boxed{v_n = 30.4 \text{ m/s} = 68.0 \text{ mph}} \tag{A.3.7}$$

A.3.8. Speed response and amplification limit. The speed sweep uses 2000 equally spaced values from 0 to 55.88 m/s, with $f = v/\lambda$. The analytic event values are placed at their exact speeds.

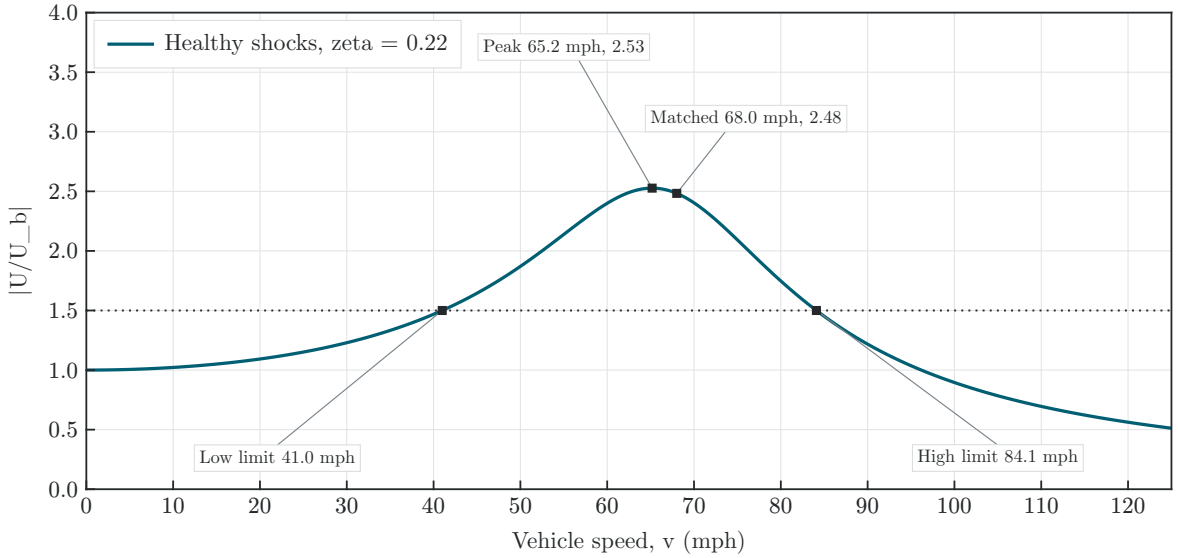


Figure 8: *A.3.8. Healthy-shock absolute body-to-road displacement FRF versus vehicle speed. The dotted line is the 1.5 amplification threshold; square markers show the peak, matched-frequency point, and both crossings.*

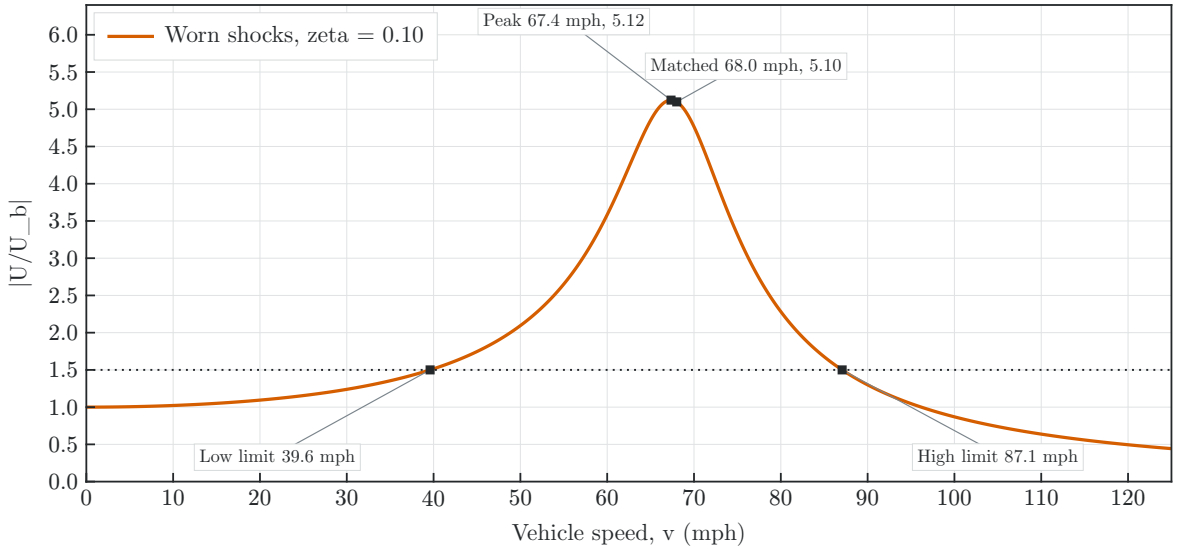


Figure 9: *A.3.10. Worn-shock absolute body-to-road displacement FRF versus vehicle speed. The dotted line is the 1.5 amplification threshold; square markers show the peak, matched-frequency point, and both crossings.*

(a) Matched-frequency event. The speed conversion places f_n at $v_n = 68.0114$ mph, and the ordinate is $|H_n| = 2.48300$. These agree with A.3.6 and A.3.7.

(b) True peak event. Differentiating $|H|^2$ with respect to $q = r^2$ gives the stationary condition

$$4\zeta^2 q^2 + 2q - 2 = 0.$$

The positive root $q = [-1 + \sqrt{1 + 8\zeta^2}]/(4\zeta^2)$ for $\zeta = 0.220$ gives $r_{\mathrm{cr}} = \sqrt{q} = 0.958311$, $f_{\mathrm{cr}} = 0.910396$ Hz, $v_{\mathrm{cr}} = 29.1363$ m/s $= 65.176$ mph, and $|H_{\max}| = 2.52688$. This peak precedes the matched-frequency point.

(c) Threshold crossings. Setting $q = r^2$ and $L = 1.5$ reduces $|H| = L$ to

$$L^2 q^2 + \left[L^2(4\zeta^2 - 2) - 4\zeta^2\right] q + (L^2 - 1) = 0.$$

There are two real positive crossings in the speed range for each shock condition. The first crossing is the limit encountered when accelerating from rest; the second is the speed above which amplification falls below 1.5 again.

$$\boxed{\begin{aligned} \zeta = 0.220: \quad & v_{\text{low}} = 18.33 \text{ m/s} = 41.0 \text{ mph}, \\ & v_{\text{high}} = 37.6 \text{ m/s} = 84.1 \text{ mph}, \\ \zeta = 0.100: \quad & v_{\text{low}} = 17.70 \text{ m/s} = 39.6 \text{ mph}, \\ & v_{\text{high}} = 38.9 \text{ m/s} = 87.1 \text{ mph}. \end{aligned}} \tag{A.3.8}$$

A.3.9. Wavelength for a 35 mph matched speed. To make the road frequency equal f_n at $v_n = 35$ mph $= 15.6464$ m/s, choose

$$\lambda_{\text{new}} = \frac{v_n}{f_n} = \frac{15.6464 \text{ m/s}}{0.950 \text{ Hz}} = 16.4699 \text{ m}.$$

$$\boxed{\lambda_{\text{new}} = 16.47 \text{ m} = 54.0 \text{ ft}} \tag{A.3.9}$$

A.3.10. Worn shocks. Repeating items (5)–(8) with $\zeta = 0.100$ retains the same matched frequency $f_n = 0.950$ Hz and matched speed $v_n = 68.0114$ mph, but increases the matched response to

$$|H_n| = \sqrt{1 + \frac{1}{(0.200)^2}} = 5.09902.$$

The true peak is $|H|_{\max} = 5.12277$ at $f_{\text{cr}} = 0.940818$ Hz and $v_{\text{cr}} = 30.1099$ m/s $= 67.3540$ mph. The table compares both shock conditions. Since $|H|^2$ increases with damping ratio for $r > \sqrt{2}$, the healthy response exceeds the worn response at high frequency, even though lower damping produces a much larger resonance peak.

Shock condition	$\lvert H_n\rvert$	f_{cr} (Hz)	Peak $\lvert H\rvert$	v_{cr} (mph)
Healthy, $\zeta = 0.220$	2.48	0.910	2.53	65.2
Worn, $\zeta = 0.100$	5.10	0.941	5.12	67.4

$$\boxed{|H_n|_{\text{worn}} = 5.10, \quad |H|_{\text{max,worn}} = 5.12 \text{ at } 0.941 \text{ Hz}} \tag{A.3.10}$$

MATLAB code. *MATLAB calculations and direct plots for A.3.*

```
% A.3. Car motion under road displacement excitation.
clear; clc; close all;
a = 2.5*0.0254; L = 105*0.3048;  % road amplitude and wavelength, m
fn = 0.95; z = [0.22 0.10];      % natural frequency, Hz; damping ratios
vm = 125*0.44704; N = 2000;      % maximum speed, m/s; sample count
b = 1.5;                         % amplification limit
C = [0 95 115; 213 94 0]/255;     % HW01 teal and orange
disp(table(a,L,fn,z(1),z(2),vm,N,b, ...
    'VariableNames',{'a','L','fn','z1','z2','vm','N','b'}));

% A.3.2--4. Road profile and excitation frequency.
g = 2*pi/L; fm = vm/L;
x = linspace(0,2*L,1001);
f = linspace(0,fm,N); v = f*L;
r = f/fn;
h = @(r,z) sqrt((1+(2*z*r).^2)./((1-r.^2).^2+(2*z*r).^2));
fprintf('Wavenumber = %.6g rad/m; maximum frequency = %.6g Hz\n',g,fm);
figure;
plot(x,a*cos(g*x),'Color',C(1,:),'LineWidth',1.3);
xlabel('Distance, x (m)'); ylabel('Road displacement, u_b (m)');
grid on; box on;

% A.3.5--10. Healthy and worn shocks.
vn = fn*L; Ln = 35*0.44704/fn;
fprintf('Matched speed = %.6g m/s = %.6g mph\n',vn,vn/0.44704);
fprintf('Wavelength for 35 mph matched speed = %.6g m\n',Ln);
figure; hold on;
plot(f,h(r,z(1)),'Color',C(1,:),'LineWidth',1.3);
plot(f,h(r,z(2)),'--','Color',C(2,:),'LineWidth',1.3);
xlabel('Road frequency, f (Hz)'); ylabel('|U/U_b|');
legend('Healthy shocks','Worn shocks'); grid on; box on;

for j = 1:2
    H = h(r,z(j));
    rp = sqrt(2/(1+sqrt(1+8*z(j)^2)));
    fp = rp*fn; vp = fp*L; hp = h(rp,z(j)); hn = h(1,z(j));
    q = roots([b^2, b^2*(4*z(j)^2-2)-4*z(j)^2, b^2-1]);
    vl = sort(sqrt(q)*fn*L);
    fprintf('\nzeta = %.2f\n',z(j));
    fprintf('Matched magnitude = %.6g\n',hn);
    fprintf('Peak = %.6g at %.6g Hz and %.6g mph\n',hp,fp,vp/0.44704);
    fprintf('Limit speeds = %.6g and %.6g mph\n', ...
        vl(1)/0.44704,vl(2)/0.44704);
    figure;
    plot(v/0.44704,H,'Color',C(j,:),'LineWidth',1.3);
    yline(b,':k');
    xlabel('Vehicle speed, v (mph)'); ylabel('|U/U_b|');
    legend(sprintf('zeta = %.2f',z(j)),'Limit');
    grid on; box on;
end
```

A.4 Standing Waves

(1) Follow the notes to develop the standing-wave concept:

- **(a)** Write the complex-exponential expressions for two identical harmonic waves of amplitude $1/2$ propagating in opposite directions.
- **(b)** Add the two waves and write the resulting complex-exponential expression.
- **(c)** Use Euler's formula to arrive at a trigonometric expression.

(2) Sketch a standing wave and identify the nodes and antinodes (bows).

(3) Assume wavespeed $c = 340\,\mathrm{m/s}$ and frequency $f = 2.9\,\mathrm{kHz}$. Calculate:

- **(a)** Rad/sec frequency ω.
- **(b)** Wavenumber γ.
- **(c)** Wavelength λ.
- **(d)** Distance Δx between standing wave nodes.

(4) Graduate work. Explain intuitively the analogy between standing waves and the vibration of a taut string.

Solution to A.4

The course notes use the time convention $e^{-i\omega t}$. With that convention, a forward wave travels in the positive x direction and a backward wave travels in the negative x direction. For A.4.1.a, the two waves with amplitude $1/2$ are

$$\boxed{u^+(x,t) = \frac{1}{2}e^{i(\gamma x - \omega t)}, \qquad u^-(x,t) = \frac{1}{2}e^{-i(\gamma x + \omega t)}} \tag{A.4.1.a}$$

Adding these waves and factoring out their common time dependence gives the complex standing-wave expression

$$\boxed{\begin{aligned} u(x,t) &= u^+(x,t) + u^-(x,t) \\ &= \frac{1}{2}\left(e^{i\gamma x} + e^{-i\gamma x}\right) e^{-i\omega t} \\ &= \cos(\gamma x) e^{-i\omega t}. \end{aligned}} \tag{A.4.1.b}$$

Euler's formula, $\cos(\gamma x) = \frac{1}{2}(e^{i\gamma x} + e^{-i\gamma x})$, gives the real trigonometric displacement

$$\boxed{u_{\text{phys}}(x,t) = \Re\{u(x,t)\} = \cos(\gamma x)\cos(\omega t)} \tag{A.4.1.c}$$

Its spatial shape stays fixed while its amplitude oscillates in time. The nodes occur where the cosine factor vanishes, and the bows, or antinodes, occur where its magnitude is one.

A.4.2. Nodes and antinodes. The first node is at $x = \lambda/4$ for the cosine phase chosen above. Subsequent nodes are separated by $\lambda/2$, while antinodes lie halfway between adjacent nodes. The figure shows one wavelength and marks both fixed nodes and antinodes.

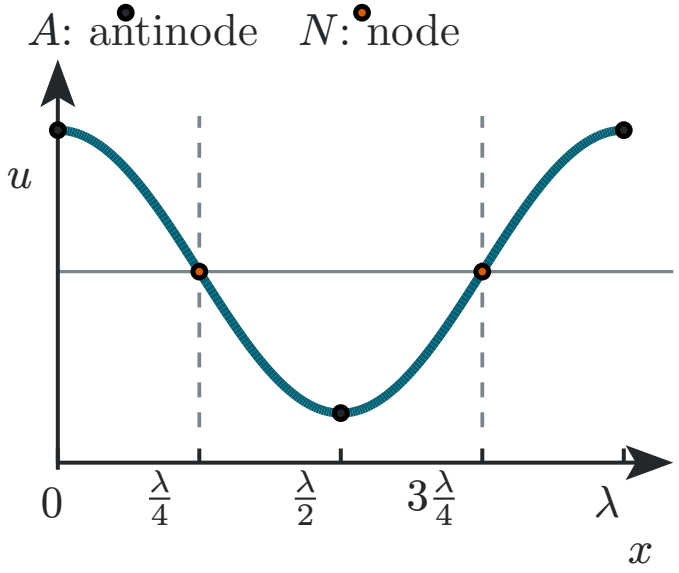


Figure 10: *A.4.2. Standing-wave displacement shape at an instant. Nodes remain fixed; antinodes reach the maximum displacement.*

For A.4.3, the input data are

Quantity	Value	Unit
c	340	m/s
f	2.9	kHz

The frequency and wavespeed give

$$\omega = 2\pi f, \qquad \gamma = \frac{\omega}{c} = \frac{2\pi}{\lambda}, \qquad \lambda = \frac{c}{f}.$$

With $c = 340$ m/s and $f = 2.9$ kHz $= 2900$ Hz, the angular frequency and wavenumber are

$$\omega = 2\pi(2900 \text{ Hz}) = 18221.2374 \text{ rad/s},$$
$$\gamma = \frac{18221.2374 \text{ rad/s}}{340 \text{ m/s}} = 53.5919 \text{ rad/m}.$$

These values give the wavelength and node spacing

$$\lambda = \frac{340 \text{ m/s}}{2900 \text{ Hz}} = 0.117241 \text{ m}, \qquad \Delta x = \frac{\pi}{\gamma} = \frac{\lambda}{2} = 0.0586207 \text{ m}.$$

The resulting values are

$$\begin{aligned} \omega &= 1.822 \times 10^4 \text{ rad/s} && \text{(A.4.3.a)}, \\ \gamma &= 53.6 \text{ rad/m} && \text{(A.4.3.b)}, \\ \lambda &= 0.1172 \text{ m} && \text{(A.4.3.c)}, \\ \Delta x &= 0.0586 \text{ m} && \text{(A.4.3.d)}. \end{aligned}$$

A.4.4. Taut-string analogy. A small transverse disturbance on a taut string travels along the string, reflects from a fixed end, and overlaps the returning wave. When the incident and reflected waves have the same frequency and amplitude, their superposition creates fixed points of zero displacement and intervening bows that oscillate in place. Adjacent bows move in opposite directions. The string therefore vibrates as a standing mode even though each component wave propagates. A string fixed at both ends must have a node at each end, so its free standing modes require an integer number of half wavelengths, $L = n\lambda/2$ for $n = 1, 2, 3, \ldots$. The string tension and mass per unit length set the wavespeed and therefore determine the allowed frequencies together with these boundary conditions.

MATLAB code. *MATLAB calculations and standing-wave plot for A.4.*

```
% Standing waves from two equal waves traveling in opposite directions.
clear; clc;

c = 340;                % wave speed (m/s)
f = 2.9e3;              % frequency (Hz)
w = 2*pi*f;             % angular frequency (rad/s)
g = w/c;                % wavenumber (rad/m)
lam = c/f;              % wavelength (m)
dx = pi/g;              % distance between adjacent nodes (m)

fprintf('omega = %.6g rad/s\n', w);
fprintf('gamma = %.6g rad/m\n', g);
fprintf('lambda = %.6g m\n', lam);
fprintf('node spacing = %.6g m\n', dx);

% Show one spatial wavelength at three instants in the oscillation.
x = linspace(0, lam, 500);
t = [0, 1/8, 1/2]/f;
u = cos(g*x(:))*cos(w*t);
C = [0 95 115; 213 94 0; 37 40 42]/255; % HW01 colors

figure('Color','w');
plot(x, u(:,1), 'Color', C(1,:), 'LineWidth', 1.4);
hold on;
plot(x, u(:,2), 'Color', C(2,:), 'LineWidth', 1.4);
plot(x, u(:,3), 'Color', C(3,:), 'LineWidth', 1.4);
grid on;
xlabel('Position, x (m)');
ylabel('Displacement, u (normalized)');
xlim([0 lam]);
legend('t = 0', 't = T/8', 't = T/2', 'Location', 'best');
```

A.5 Musical Notes

(1) Find the Hz frequency of note C_5.

(2) Find what musical note corresponds to 246.94 Hz.

(3) What is the lowest frequency on the scale of equal temperament and what musical note does it correspond to?

(4) What is the commonly accepted frequency range of human hearing?

(5) What is the highest frequency on the scale of equal temperament and what musical note does it correspond to?

(6) Graduate work. Explain the scale of equal temperament by answering items (a)–(c).

- **(a)** What is the organizing principle of the scale of equal temperament.
- **(b)** How many equal-temperament entries are in an octave?
- **(c)** What is the ratio between two adjacent entries on the scale of equal temperament?

Solution to A.5

The course notes anchor the equal-temperament scale at $A_4 = 440$ Hz and divide each octave into twelve equal semitone intervals. If n is the number of semitones above or below A_4, the frequency is

$$f_n = 440\,2^{n/12}\ \text{Hz}.$$

The input values and the note range shown in the course-note chart are

Quantity	Value	Unit
A_4	440	Hz
f	246.94	Hz
Chart range	C_0–C_{10}	notes

The chart appears on page 90 of the course notes.

For A.5.1, C_5 is three semitones above A_4. For A.5.2, the frequency-to-note relation gives the semitone offset

$$n = 12\log_2\left(\frac{246.94}{440}\right) = -10.0001 \approx -10,$$

so the supplied frequency corresponds to B_3, ten semitones below A_4.

$$f_{C_5} = 440\,2^{3/12} = 523.251131\ \text{Hz} \approx \boxed{523\ \text{Hz}} \qquad \text{(A.5.1)},$$

$$f_{B_3} = 440\,2^{-10/12} = 246.941651\ \text{Hz} \approx 246.94\ \text{Hz},$$

$$246.94\ \text{Hz} \longleftrightarrow \boxed{B_3} \qquad \text{(A.5.2)}.$$

The course-note chart begins at C_0 and ends at C_{10}. Applying the same equation gives its endpoints. Equal temperament itself continues by octave in both directions, so the mathematical scale has no absolute lowest or highest frequency.

$$f_{C_0} = 440\,2^{-57/12} = 16.351598\ \text{Hz} \approx \boxed{16.35\ \text{Hz at } C_0} \qquad \text{(A.5.3, chart minimum)},$$

$$f_{C_{10}} = 440\,2^{63/12} = 16744.036179\ \text{Hz} \approx \boxed{16.74\ \text{kHz at } C_{10}} \qquad \text{(A.5.5, chart maximum)}.$$

The commonly accepted hearing range is approximately 20 Hz to 20 kHz [2].

$$\boxed{20\ \text{Hz} \le f \le 20\ \text{kHz}} \qquad \text{(A.5.4)}$$

Equal temperament spaces the twelve semitone steps uniformly on a logarithmic frequency scale. Each successive entry is therefore the previous frequency multiplied by the same factor, and twelve such steps double the frequency to reach the same note in the next octave.

$$\boxed{\begin{aligned}&\text{entries per octave} = 12,\\ &\frac{f_{n+1}}{f_n} = 2^{1/12} \approx 1.059,\\ &\left(2^{1/12}\right)^{12} = 2.\end{aligned}} \qquad \text{(A.5.6)}$$

MATLAB code. *MATLAB calculations for A.5.*

```
% A.5. Equal-temperament frequencies and note names.
clear; clc;

A4 = 440;                  % tuning reference, Hz
f = 246.94;                % supplied frequency, Hz
n = round(12*log2(f/A4));  % nearest semitone from A4

C5 = A4*2^(3/12);          % C5, three semitones above A4
B3 = A4*2^(-10/12);        % B3, ten semitones below A4
C0 = A4*2^(-57/12);        % lowest note shown in course-note chart
C10 = A4*2^(63/12);        % highest note shown in course-note chart
r = 2^(1/12);              % adjacent semitone frequency ratio

fprintf('C5 = %.6g Hz\n', C5);
fprintf('%.2f Hz is nearest to B3 (semitone offset from A4 = %d)\n', f, n);
fprintf('Course-note chart span: C0 = %.6g Hz to C10 = %.6g Hz\n', C0, C10);
fprintf('Nominal human hearing range = 20 Hz to 20 kHz\n');
fprintf('Equal temperament: 12 entries per octave; adjacent ratio = %.6g\n', r);
```

A.6 Musical Instruments

(1) List at least three methods to produce sound in a string instrument.

(2) List the two different methods to produce a sound in wind instruments.

(3) List string instruments that use a hammer hit to produce sound.

(4) List the three major categories of musical instruments. For each category, complete items (a) and (b).

- **(a)** Describe how the instrument works.
- **(b)** Explain what physical principles are used to produce sound.

(5) List at least five common string instruments.

Solution to A.6

The course notes classify musical instruments into string, percussion, and wind instruments (p. 88). In each case, a mechanical oscillator drives pressure fluctuations in the surrounding air. The oscillator and its boundary conditions determine the frequency content, while the instrument body or air column couples that motion to the listener.

1. Methods of exciting a string. A string can be plucked, bowed, or struck with a hammer. Plucking gives the string an initial displacement and releases it, bowing supplies repeated frictional excitation, and hammering supplies a short impact. In all three cases, the string vibrates in standing-wave modes and transfers energy through the bridge to a soundboard or instrument body, which radiates the sound more effectively into the air.

$$\boxed{\text{Pluck, bow, or strike the string.}} \qquad \text{(A.6.1)}$$

2. Methods of exciting a wind instrument. The two broad methods are edge-tone excitation and reed excitation. In an edge-tone instrument, a focused air jet alternates across a sharp opening and excites the air column, as in a flute. In a reed instrument, airflow makes a flexible reed or the player's lips open and close the passage periodically, as in a clarinet, oboe, or trumpet. The resulting pressure disturbance couples to resonant modes of the air column [7, 6].

$$\boxed{\begin{array}{c}\text{Edge-tone excitation by an air jet, and}\\ \text{reed or lip excitation by a periodically opening passage.}\end{array}} \qquad \text{(A.6.2)}$$

3. String instruments excited by a hammer. Examples include the piano and hammered dulcimer. The course notes describe the hammered dulcimer as a percussion-stringed instrument whose strings are struck by small hammers and whose resonant soundboard radiates the resulting vibration (pp. 170–171).

$$\boxed{\text{Piano and hammered dulcimer.}} \qquad \text{(A.6.3)}$$

4. Major categories of musical instruments. The three categories used in the course notes are string, percussion, and wind instruments.

String instruments. A stretched string is excited by plucking, bowing, or impact. The string supports transverse standing waves, and its natural frequencies depend on its length, tension, and mass per unit length. The bridge and soundboard couple the string motion to the air and strengthen the radiated sound.

$$\boxed{\text{String: vibrating stretched strings and their standing-wave modes.}} \qquad \text{(A.6.4a)}$$

Percussion instruments. A membrane, bar, plate, or instrument body is struck, shaken, or scraped. The short interaction supplies an impulse that excites the object's natural modes. Its mass, stiffness, geometry, and boundary conditions determine the modal frequencies, while damping controls the decay of the sound.

Percussion: impact or motion excites structural vibration modes. (A.6.4b)

Wind instruments. A jet, reed, or vibrating lips excites an air column inside a tube. The pressure and particle-displacement waves reflect from the tube boundaries and form standing modes. Open and closed ends impose different pressure and displacement conditions, so the tube length, openings, valves, and bore determine the resonant frequencies. The course notes use the vibrating reed as a wind-instrument application of cantilever-beam vibration (p. 211).

Wind: an air-column resonance driven by an edge jet, reed, or lips. (A.6.4c)

5. Common string instruments. Common examples are the violin, viola, cello, double bass, guitar, harp, piano, and hammered dulcimer. The piano and hammered dulcimer are string instruments even though their strings are excited by hammers rather than by fingers or a bow.

Violin, viola, cello, double bass, guitar, harp, piano, and hammered dulcimer. (A.6.5)

2 Fluid Waves and Vibration

B.1 Analyze the Wave Equation in Fluids

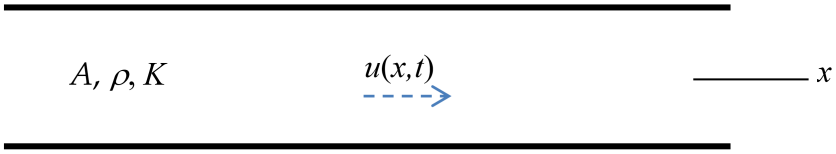


Figure 11: *Fluid waves in an infinite tube.*

Assume an infinite tube of cross section A containing fluid of density ρ and bulk modulus K (Figure 11). Develop the wave equation in fluids.

Do the following.

(1) Assume that the fluid in the tube undergoes time-varying axial displacement $u(x,t)$ under the influence of a pressure disturbance $p(x,t)$.

(2) Consider an infinitesimal element dx and draw the displacements and pressures on both of its sides.

(3) Draw FBD.

(4) Apply NLM and deduce relationship between pressure p, density ρ and acceleration $\ddot{u}$.

(5) Apply fluid-elasticity equation and express the pressure $p(x,t)$ in terms of displacement $u(x,t)$ and bulk modulus K.

(6) Substitute (5) into (4) and deduce the relationship between the second derivative with respect to space, u'', and the second derivative with respect to time, $\ddot{u}$.

(7) Define the wavespeed c.

(8) Substitute (7) into (6) and deduce the wave equation.

Solution to B.1

The derivation follows the one-dimensional, small-motion model used in the course notes (pp. 103–105). The tube has rigid walls and constant cross-sectional area A. The fluid is inviscid, has uniform equilibrium density ρ, and has constant bulk modulus K. There is no mean flow. Subscripts x and t denote partial derivatives with respect to position and time. The pressure variable below is the acoustic pressure disturbance, so the total pressure is $P(x,t) = P_0 + p(x,t)$, where the uniform equilibrium pressure P_0 has zero spatial gradient.

1. Displacement and pressure disturbance. Let $u(x,t)$ be the small axial displacement of a fluid particle whose equilibrium position is x. The associated pressure disturbance is $p(x,t)$. The element initially occupying $[x, x+dx]$ has left and right displacements $u(x,t)$ and $u(x+dx,t)$, respectively. A first-order Taylor expansion gives

$$u(x+dx,t) = u(x,t) + u_x(x,t)dx, \qquad p(x+dx,t) = p(x,t) + p_x(x,t)dx.$$

The longitudinal strain is therefore u_x. The symbol p denotes the perturbation from equilibrium; the constant P_0 cancels from the force balance.

2–3. Infinitesimal element and free-body diagram. The element and its pressure forces are shown below. The two displacement arrows describe the particle motion at the element faces. The force arrows in the lower panel are pressure forces. Pressure on the left face acts in the positive x direction, while pressure on the right face acts in the negative x direction.

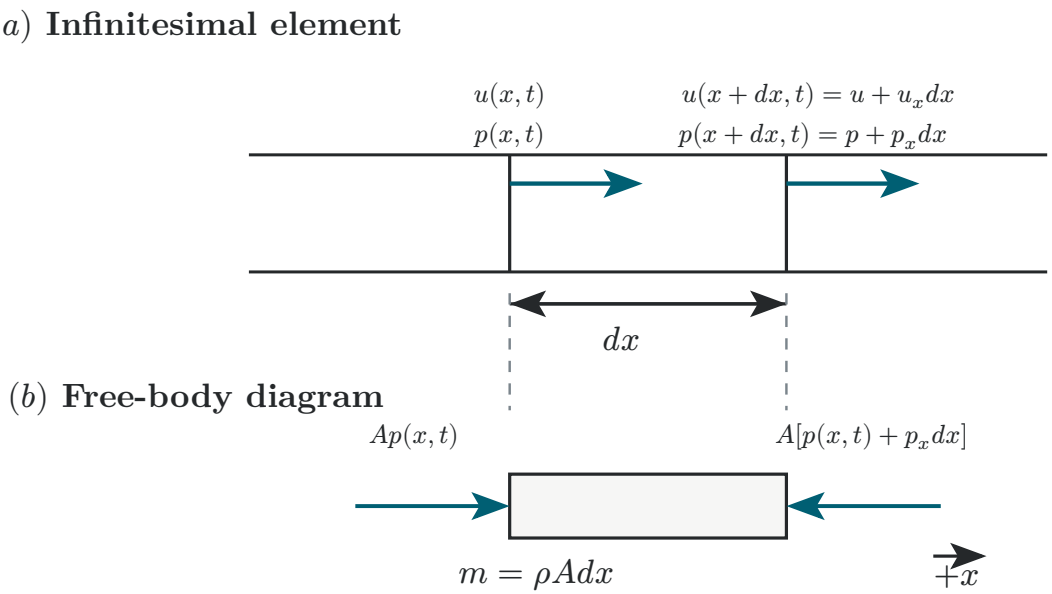


Figure 12: *Infinitesimal fluid element and its free-body diagram. The upper panel shows the pressure disturbance and axial displacement at each face. The lower panel shows the pressure forces acting on the isolated element.*

4. Newton's law of motion. The pressure force on the left face is $Ap(x,t)$ in the positive x direction. The pressure force on the right face is $-A[p(x,t) + p_x(x,t)dx]$. Thus the net force is

$$F_x = Ap - A(p + p_x dx) = -Ap_x dx.$$

The element mass is $m = \rho A dx$, and its axial acceleration is u_{tt}. Newton's law gives

$$-Ap_x dx = (\rho A dx) u_{tt}.$$

After canceling Adx, the pressure–acceleration relation is

$$\boxed{p_x = -\rho u_{tt}} \qquad \text{(B.1.4)}$$

The sign follows directly from the fact that pressure decreases in the direction of positive acceleration.

5. Fluid elasticity. The element volume is $V = Adx$. Its first-order volume change is

$$\Delta V = A\big[u(x + dx, t) - u(x, t)\big] = Au_x dx.$$

The fluid-elasticity relation is $\Delta P = -K\Delta V/V$. The equilibrium pressure is constant, so $\Delta P = p$. Therefore,

$$p = -K\frac{Au_x dx}{Adx}, \qquad \boxed{p(x,t) = -Ku_x(x,t)} \qquad \text{(B.1.5)}$$

Compression has $u_x < 0$, which produces a positive pressure disturbance, consistent with the sign convention.

6. Relationship between spatial and temporal derivatives. Differentiating the elasticity relation with respect to x gives $p_x = -Ku_{xx}$. Equating this result to the Newton-law relation from item 4 gives

$$-Ku_{xx} = -\rho u_{tt}, \qquad \boxed{Ku_{xx} = \rho u_{tt}} \qquad \text{(B.1.6)}$$

7. Wavespeed. Define the fluid wavespeed, also called the sound speed, by

$$\boxed{c = \sqrt{\frac{K}{\rho}}} \quad \Longleftrightarrow \quad \boxed{c^2 = \frac{K}{\rho}} \qquad \text{(B.1.7)}$$

8. Wave equation. Dividing the relation from item 6 by ρ and substituting $K/\rho = c^2$ yields the one-dimensional wave equation in the fluid,

$$\boxed{u_{tt} = c^2 u_{xx}} \quad \text{or equivalently} \quad \boxed{u_{xx} - \frac{1}{c^2}u_{tt} = 0} \qquad \text{(B.1.8)}$$

This equation describes small axial disturbances in the uniform inviscid fluid under the stated rigid-tube assumptions.

B.2 Analyze and Solve the Wave Equation

Recall the results of Problem B.1.

Do the following.

(1) Recall the wave equation.

(2) Assume separation of time and space behaviors into $U(x)$ and $T(t)$ and write $u(x,t)$ as a function of $U(x)$ and $T(t)$.

(3) Perform the appropriate differentiations with respect to space and time.

(4) Assume vibratory time behavior with angular frequency ω using a complex exponential formula for $T(t)$.

(5) Substitute (4) into (3) and write expressions for u'' and $\ddot{u}$.

(6) Substitute (5) into the wave equation (1) and express it only in terms of $U(x)$, c, ω.

(7) Define wavenumber γ.

(8) Substitute (7) into (6) and obtain a linear ordinary differential equation (ODE) in terms of $U(x)$, $U''(x)$, and γ.

(9) Assume exponential solution of the form $U(x) = e^{sx}$.

(10) Substitute (9) into (8) and obtain the characteristic equation.

(11) Solve the characteristic equation (10) and obtain the roots s_1, s_2.

(12) Substitute (11) into (9) and write the complete solution in terms of constants A_1, A_2 and complex exponentials of s_1, s_2.

(13) Add the vibratory time behavior from (4) to obtain the solution in terms of complex exponentials depending on both γ and ω.

(14) Use Euler's formulas to rewrite (13) in terms of new constants C_1 and C_2, trigonometric functions depending on γ, and a complex exponential depending on ω.

Solution to B.2

1. Wave equation. The wave equation obtained in Problem B.1 is

$$\boxed{u_{tt} = c^2 u_{xx}} \qquad \text{(B.2.1)}$$

where $u(x,t)$ is the axial displacement and c is the wavespeed.

2. Separation of variables. Assume that the spatial and temporal behaviors can be separated as

$$u(x,t) = U(x)T(t).$$

3. Differentiation. The required derivatives are

$$u_{xx} = U''(x)T(t), \qquad u_{tt} = U(x)T''(t).$$

4. Harmonic time dependence. For vibratory motion with angular frequency ω, use the course-note convention

$$T(t) = e^{-i\omega t}.$$

5. Harmonic derivatives. Substitution of this time behavior into the separated solution and its derivatives gives

$$u(x,t) = U(x)e^{-i\omega t}, \qquad u_{xx} = U''(x)e^{-i\omega t}, \qquad u_{tt} = -\omega^2 U(x)e^{-i\omega t}.$$

6. Spatial equation before defining the wavenumber. Substitution into the wave equation gives

$$c^2 U''(x)e^{-i\omega t} = -\omega^2 U(x)e^{-i\omega t}.$$

After canceling the nonzero time factor,

$$\boxed{c^2 U''(x) + \omega^2 U(x) = 0} \qquad \text{(B.2.6)}$$

7. Wavenumber. Define the wavenumber by

$$\boxed{\gamma = \frac{\omega}{c}} \qquad \text{(B.2.7)}$$

8. Spatial ODE. For $\omega > 0$ and $c > 0$, dividing the equation from item 6 by c^2 gives

$$\boxed{U''(x) + \gamma^2 U(x) = 0} \qquad \text{(B.2.8)}$$

9. Exponential trial solution. Assume

$$U(x) = e^{sx} \qquad \text{(B.2.9)}$$

10. Characteristic equation. Then $U''(x) = s^2 e^{sx} = s^2 U(x)$. Substitution into the spatial ODE gives

$$\boxed{s^2 + \gamma^2 = 0} \qquad \text{(B.2.10)}$$

11. Characteristic roots. Solving the characteristic equation gives

$$\boxed{s_1 = +i\gamma, \qquad s_2 = -i\gamma} \qquad \text{(B.2.11)}$$

12. Complete spatial solution. The two independent exponential solutions give the linear combination

$$\boxed{U(x) = A_1 e^{i\gamma x} + A_2 e^{-i\gamma x}} \qquad \text{(B.2.12)}$$

where A_1 and A_2 are arbitrary complex constants.

13. Complete space–time solution. Restoring the harmonic time factor gives

$$\boxed{u(x,t) = A_1 e^{i(\gamma x - \omega t)} + A_2 e^{-i(\gamma x + \omega t)}} \qquad \text{(B.2.13)}$$

The first term has constant phase when $x - ct$ is constant and therefore travels in the positive x direction. The second term has constant phase when $x + ct$ is constant and travels in the negative x direction.

14. Trigonometric form and physical displacement. Euler's formulas give

$$\begin{aligned} U(x) &= A_1(\cos\gamma x + i\sin\gamma x) + A_2(\cos\gamma x - i\sin\gamma x) \\ &= \big[i(A_1 - A_2)\big]\sin\gamma x + (A_1 + A_2)\cos\gamma x. \end{aligned}$$

Define

$$C_1 = i(A_1 - A_2), \qquad C_2 = A_1 + A_2.$$

The solution can therefore be written as

$$\boxed{U(x) = C_1 \sin\gamma x + C_2 \cos\gamma x}$$

and

$$\boxed{u(x,t) = \big(C_1 \sin\gamma x + C_2 \cos\gamma x\big) e^{-i\omega t}} \qquad \text{(B.2.14)}$$

The measurable displacement is the real part,

$$\boxed{u_{\text{phys}}(x,t) = \Re\{u(x,t)\}.}$$

The constants and frequency are selected by initial or boundary data when those data are specified. Problem B.2 considers an infinite tube, so no boundary conditions are imposed and no discrete eigenvalue condition is introduced; the result is the monochromatic wave family parameterized by ω and the associated $\gamma = \omega/c$.

B.3 Analyze Air Vibration in a Tube Closed at Both Ends

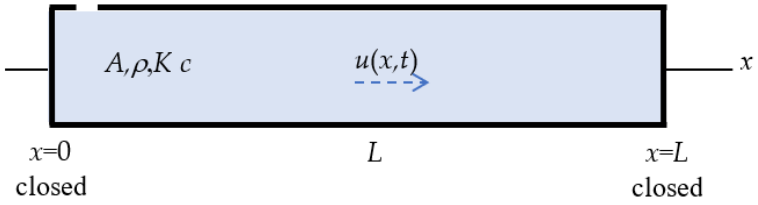


Figure 13: *Air vibration in a tube of finite length L closed at both ends.*

Assume a tube of finite length L and cross section A containing fluid of density ρ, bulk modulus K, and wavespeed c (Figure 13). The tube is closed at both ends. The air in the tube undergoes harmonic axial displacement $u(x,t)$ with frequency ω.

Do the following.

(1) Write the boundary conditions at the two ends of the tube in terms of displacement $u(x,t)$.

(2) Recall the harmonic solution of the wave equation consisting of the multiplication of two terms:

- **(a)** First term: a function $U(x)$ consisting of trigonometric functions depending on wavenumber γ, space variable x, and constants C_1, C_2.
- **(b)** Second term: a complex exponential depending on frequency ω and time variable t.

(3) Substitute (2) into (1) and write the boundary conditions at the two ends of the tube in terms of unknown constants C_1, C_2.

(4) Solve (3) for C_2 and obtain an equation for C_1.

(5) Discuss the trivial solution.

(6) Choose to search for the nontrivial solution and define the equation that needs to be solved to find a nontrivial solution.

(7) Introduce the notation $z = \gamma L$ into (6) and write the eigenvalue equation in terms of z.

(8) Solve the eigenvalue equation and write the general expression for the eigenvalues z_j, where $j = 1, 2, 3, \ldots$.

(9) Use (7) to write the general expression for the wavenumber γ in terms of j and length L.

(10) Recall from HW01 the definition of wavenumber γ in terms of frequency ω and wavespeed c.

(11) Solve (10) for frequency ω in terms of wavenumber γ and wavespeed c.

(12) Use (9) and (11) to write the general expression of the natural frequency ω_j in terms of j, wavespeed c, and tube length L.

(13) Use (12) to write the general expression of the Hz frequency f in terms of j, wavespeed c, and tube length L.

(14) Explain the meaning of fundamental frequency and overtones.

(15) Explain the meaning of fundamental harmonic and higher harmonics.

(16) Recall the relationship between wavelength λ and wavenumber γ.

(17) Substitute (9) into (16) to write the general expression for the wavelength λ_j in terms of j and tube length L.

(18) Use (17) to explain which wavelength multiples could accommodate vibratory standing waves in a tube of length L closed at both ends.

(19) Recall $U(x)$ from (2)(a) and use (4) to eliminate C_2.

(20) Set $C_1 = 1$ in item (19) and write the general expression for the displacement modeshape $U_j(x)$ in terms of the wavenumber γ_j.

(21) Write the general expression of the pressure modeshapes.

Solution to B.3

The analysis considers a uniform tube of length L and constant cross-sectional area A closed rigidly at both ends ($x = 0$ and $x = L$), as shown in Figure 13. The tube contains an inviscid fluid of equilibrium density ρ, bulk modulus K, and acoustic sound speed $c = \sqrt{K/\rho}$. Small-amplitude axial vibrations $u(x, t)$ satisfy the one-dimensional wave equation established in Problems B.1 and B.2.

1. Boundary conditions in terms of displacement. At a rigid, impermeable wall, fluid particles cannot move into or out of the boundary. Therefore, the axial displacement must vanish at both closed ends for all time:

$$\boxed{u(0,t) = 0} \qquad \text{(left closed end, B.3.1.a)}$$

$$\boxed{u(L,t) = 0} \qquad \text{(right closed end, B.3.1.b)}$$

2. Harmonic solution of the wave equation. From Problem B.2 (item 14), the monochromatic standing-wave solution consists of the product of a spatial function $U(x)$ and a temporal harmonic factor $T(t)$:

(a) First term (spatial distribution):

$$\boxed{U(x) = C_1 \sin(\gamma x) + C_2 \cos(\gamma x)} \qquad \text{(B.3.2.a)}$$

where γ is the wavenumber and C_1, C_2 are arbitrary constants.

(b) Second term (temporal harmonic oscillation):

$$\boxed{T(t) = e^{-i\omega t}} \qquad \text{(B.3.2.b)}$$

where ω is the angular frequency.

The complete displacement solution is therefore

$$\boxed{u(x,t) = U(x)T(t) = \big(C_1 \sin \gamma x + C_2 \cos \gamma x\big)e^{-i\omega t}} \qquad \text{(B.3.2.c)}$$

3. Boundary conditions in terms of constants C_1 and C_2. Substituting the harmonic solution into the boundary conditions at $x = 0$ and $x = L$ gives:

$$u(0,t) = \big(C_1 \sin 0 + C_2 \cos 0\big)e^{-i\omega t} = C_2 e^{-i\omega t} = 0 \qquad \text{(B.3.3.a)}$$

$$u(L,t) = \big(C_1 \sin \gamma L + C_2 \cos \gamma L\big)e^{-i\omega t} = 0 \qquad \text{(B.3.3.b)}$$

4. Determination of C_2 and equation for C_1. Because $e^{-i\omega t} \neq 0$ for all t, condition (B.3.3.a) requires

$$\boxed{C_2 = 0} \qquad \text{(B.3.4.a)}$$

Substituting $C_2 = 0$ into condition (B.3.3.b) eliminates the cosine term and yields

$$\boxed{C_1 \sin(\gamma L) = 0} \qquad \text{(B.3.4.b)}$$

5. Discussion of the trivial solution. If $C_1 = 0$, then since $C_2 = 0$, the spatial function vanishes identically: $U(x) \equiv 0$, and consequently $u(x,t) \equiv 0$ for all $x \in [0, L]$ and all t. This trivial solution represents the fluid at rest in equilibrium, in which no vibration, acoustic disturbance, or sound wave is present.

6. Nontrivial solution and governing condition. For a sustained, physically meaningful vibration to exist, we seek a nontrivial solution where the amplitude is nonzero, $C_1 \neq 0$. Dividing (B.3.4.b) by C_1 requires that the wavenumber γ satisfy

$$\boxed{\sin(\gamma L) = 0} \qquad \text{(B.3.6)}$$

When this condition holds, C_1 can take any arbitrary nonzero value. This reflects the single degree of indeterminacy characteristic of linear modal analysis, where absolute vibration amplitudes are set by external excitation or initial conditions rather than boundary equations alone.

7. Eigenvalue equation. Introduce the dimensionless parameter z defined by

$$\boxed{z = \gamma L} \qquad \text{(B.3.7.a)}$$

Substituting this definition into the condition from item 6 yields the eigenvalue equation:

$$\boxed{\sin z = 0} \qquad \text{(B.3.7.b)}$$

8. Eigenvalue solutions. The mathematical roots of $\sin z = 0$ are $z = n\pi$ for any integer n. For physical vibrations characterized by positive propagation wavenumber ($\gamma > 0$) and positive length ($L > 0$), we restrict attention to the positive integers:

$$\boxed{z_j = j\pi, \qquad j = 1, 2, 3, \ldots} \qquad \text{(B.3.8)}$$

The values $z_1 = \pi, z_2 = 2\pi, z_3 = 3\pi, \ldots$ are the eigenvalues of the closed-closed tube system.

9. General expression for the wavenumbers. Using the relation $z_j = \gamma_j L$, the modal wavenumbers are

$$\boxed{\gamma_j = \frac{z_j}{L} = \frac{j\pi}{L}, \qquad j = 1, 2, 3, \ldots} \qquad \text{(B.3.9)}$$

10. Definition of wavenumber. In acoustics and wave mechanics (HW01 and Problem B.2), the wavenumber γ relates the circular frequency ω to the sound speed c via

$$\boxed{\gamma = \frac{\omega}{c}} \qquad \text{(B.3.10)}$$

11. Angular frequency in terms of wavenumber and wavespeed. Solving (B.3.10) for ω gives

$$\boxed{\omega = \gamma c} \qquad \text{(B.3.11)}$$

12. General expression for the natural frequencies. Substituting the modal wavenumbers γ_j from item 9 into the frequency relation gives the natural angular frequencies (in rad/s):

$$\boxed{\omega_j = \gamma_j c = \frac{j\pi c}{L}, \qquad j = 1, 2, 3, \ldots} \qquad \text{(B.3.12)}$$

13. General expression for cyclic frequencies in Hz. Because $\omega = 2\pi f$, the cyclic frequency in hertz (Hz or s^{-1}) is $f = \omega/(2\pi)$. Substituting (B.3.12) yields

$$\boxed{f_j = \frac{\omega_j}{2\pi} = \frac{jc}{2L}, \qquad j = 1, 2, 3, \ldots} \qquad \text{(B.3.13)}$$

14. Meaning of fundamental frequency and overtones.

Fundamental frequency: The lowest resonant frequency of the system, obtained for the lowest modal index $j = 1$:

$$f_1 = \frac{c}{2L}.$$

It represents the principal natural pitch produced by the vibrating air column.

Overtones: Any resonant frequency higher than the fundamental frequency ($j = 2, 3, 4, \ldots$). The first overtone is $f_2 = 2f_1 = c/L$; the second overtone is $f_3 = 3f_1 = 3c/(2L)$; and in general, the n-th overtone corresponds to index $j = n + 1$ with frequency $f_{n+1} = (n + 1)f_1$.

15. Meaning of fundamental harmonic and higher harmonics.

Fundamental harmonic: A harmonic is a frequency component at an integer multiple of the fundamental frequency f_1. The fundamental harmonic (also called the first harmonic) is $f_1 = 1 \cdot f_1$.

Higher harmonics: Resonance frequencies that are integer multiples of the fundamental: the second harmonic is $2f_1$, the third harmonic is $3f_1$, etc.

Comparison for closed-closed tube: Because all positive integer multiples $j = 1, 2, 3, \ldots$ satisfy the eigenvalue condition for a tube closed at both ends, all integer harmonics are supported by the modal spectrum. Thus, the 1st overtone ($j = 2$) is the 2nd harmonic, the 2nd overtone ($j = 3$) is the 3rd harmonic, and the n-th overtone is the $(n + 1)$-th harmonic.

16. Relationship between wavelength and wavenumber. The acoustic wavelength λ and wavenumber γ are connected by

$$\boxed{\lambda = \frac{2\pi}{\gamma}} \qquad \text{(B.3.16)}$$

17. General expression for modal wavelengths. Substituting the wavenumber $\gamma_j = j\pi/L$ from item 9 into (B.3.16) gives

$$\boxed{\lambda_j = \frac{2\pi}{\gamma_j} = \frac{2\pi}{j\pi/L} = \frac{2L}{j}, \qquad j = 1, 2, 3, \ldots} \qquad \text{(B.3.17)}$$

Higher modes have progressively shorter wavelengths.

18. Wavelength multiples accommodating standing waves. Rearranging (B.3.17) in terms of the tube length L yields

$$\boxed{L = j\left(\frac{\lambda_j}{2}\right), \qquad j = 1, 2, 3, \ldots} \qquad \text{(B.3.18)}$$

This relation shows that sustained vibratory standing waves can exist inside a tube closed at both ends if and only if the length of the tube accommodates an exact integer number of half-wavelengths:

$$L = \frac{\lambda_1}{2} = 2\left(\frac{\lambda_2}{2}\right) = 3\left(\frac{\lambda_3}{2}\right) = \cdots = j\left(\frac{\lambda_j}{2}\right).$$

Because both boundaries are rigid walls where displacement must be zero (displacement nodes), every valid standing wave must place a displacement node at each end, forcing the distance between the ends to be an integral multiple of node-to-node spacing ($\lambda/2$).

19. Elimination of C_2 from spatial form. Recalling $U(x) = C_1 \sin \gamma x + C_2 \cos \gamma x$ from item 2(a) and inserting $C_2 = 0$ from item 4 gives

$$\boxed{U(x) = C_1 \sin(\gamma x)} \qquad \text{(B.3.19)}$$

20. Displacement modeshapes. Setting the arbitrary scaling factor $C_1 = 1$ and evaluating at wavenumber $\gamma_j = j\pi/L$, the general expression for the j-th displacement modeshape is

$$\boxed{U_j(x) = \sin(\gamma_j x) = \sin\left(\frac{j\pi x}{L}\right), \qquad j = 1, 2, 3, \ldots} \qquad \text{(B.3.20)}$$

At both ends, $U_j(0) = \sin(0) = 0$ and $U_j(L) = \sin(j\pi) = 0$, strictly fulfilling the closed-end boundary conditions.

21. General expression of the pressure modeshapes. From fluid elasticity (Problem B.1, Eq. B.1.5), the acoustic pressure disturbance $p(x,t)$ is related to spatial strain by

$$p(x,t) = -K u_x(x,t) = -\rho c^2 u_x(x,t).$$

For harmonic mode j, let $a_j > 0$ be the displacement amplitude in meters. Substituting $u_j(x,t) = a_j U_j(x) e^{-i\omega_j t}$ gives

$$p_j(x,t) = P_j(x) e^{-i\omega_j t}, \qquad \text{where} \quad P_j(x) = -\rho c^2 a_j U_j'(x).$$

Differentiating the displacement modeshape $U_j(x) = \sin(\gamma_j x)$ yields $U_j'(x) = \gamma_j \cos(\gamma_j x)$. Therefore, the physical pressure modeshape is

$$\boxed{P_j(x) = -\rho c^2 a_j \gamma_j \cos(\gamma_j x) = -\rho c\, \omega_j a_j \cos\left(\frac{j\pi x}{L}\right)} \qquad \text{(B.3.21.a)}$$

The peak pressure magnitude is $|P_j|_{\max} = \rho c\, \omega_j a_j$. Dividing by this positive amplitude gives the normalized pressure modeshape

$$\boxed{\bar{P}_j(x) = -\cos\left(\frac{j\pi x}{L}\right)} \qquad \text{(B.3.21.b)}$$

Changing the overall sign of a mode is allowed, but displacement and pressure must change sign together to preserve $p = -K u_x$. This result reveals the fundamental spatial phase shift of $\pi/2$ between displacement and pressure: the closed ends ($x = 0, L$) are displacement nodes ($U_j = 0$) but pressure antinodes ($|\bar{P}_j| = 1$).

B.4 Calculate Air Vibration in a Tube Closed at Both Ends

Consider a tube of length $L = 0.85\,\text{m}$ closed at both ends. Take sound speed $c = 340\,\text{m/s}$ and air density $\rho = 1.2\,\text{kg/m}^3$. Use the results of Problem B.3 to calculate the modal vibration characteristics and modeshapes for $j = 1, \ldots, N$ with $N = 5$.

(1) Display input data.

(2) Calculate and display the modal vibration characteristics:

- **(a)** Eigenvalues z_j.
- **(b)** Wavenumbers γ_j.
- **(c)** Rad/s frequencies ω_j.
- **(d)** Hz frequencies f_j.

(3) Calculate and plot the displacement modeshapes.

(4) Discuss your results.

(5) Graduate work. Calculate and plot the normalized pressure modeshapes, and compare the result with item (3).

Solution to B.4

1. Input data. The physical parameters specified for the closed-closed acoustic tube are

$$\boxed{L = 0.85\,\text{m}, \qquad c = 340\,\text{m/s}, \qquad \rho = 1.2\,\text{kg/m}^3, \qquad N = 5} \tag{B.4.1}$$

The associated medium properties are the acoustic bulk modulus and the characteristic acoustic impedance:

$$K = \rho c^2 = 1.2 \times (340)^2\,\text{Pa} = 138.72\,\text{kPa}, \qquad Z = \rho c = 1.2 \times 340\,\text{Pa}\cdot\text{s/m} = 408\,\text{rayl}.$$

2. Modal vibration characteristics. Using the analytical relations derived in Problem B.3, the modal characteristics for modes $j = 1, 2, 3, 4, 5$ are evaluated as follows:

(a) Eigenvalues z_j:

$$\boxed{z_j = j\pi \quad \Longrightarrow \quad z_1 = \pi,\ z_2 = 2\pi,\ z_3 = 3\pi,\ z_4 = 4\pi,\ z_5 = 5\pi} \tag{B.4.2.a}$$

(b) Wavenumbers γ_j:

$$\boxed{\gamma_j = \frac{z_j}{L} = \frac{j\pi}{0.85}\,\text{rad/m} \approx 3.6960\,j\,\text{rad/m}} \tag{B.4.2.b}$$

(c) Natural circular frequencies ω_j: Since $c/L = 340/0.85 = 400\,\text{s}^{-1}$,

$$\boxed{\omega_j = \gamma_j c = \frac{j\pi c}{L} = 400\pi j\,\text{rad/s} \approx 1256.64\,j\,\text{rad/s}} \tag{B.4.2.c}$$

(d) Natural frequencies f_j in Hz:

$$\boxed{f_j = \frac{\omega_j}{2\pi} = \frac{jc}{2L} = 200\,j\,\text{Hz} \quad \Longrightarrow \quad 200,\ 400,\ 600,\ 800,\ 1000\,\text{Hz}} \tag{B.4.2.d}$$

The corresponding acoustic wavelengths are $\lambda_j = 2L/j = 1.70/j\,\text{m}$. The complete modal characteristics are summarized in Table 1.

Table 1: *Modal vibration characteristics of the closed-closed air tube ($L = 0.85\,\text{m}$, $c = 340\,\text{m/s}$).*

Mode j	z_j (rad)	γ_j (rad/m)	ω_j (rad/s)	f_j (Hz)	λ_j (m)
1	$\pi \approx 3.1416$	3.6960	$400\pi \approx 1256.64$	200	1.7000
2	$2\pi \approx 6.2832$	7.3920	$800\pi \approx 2513.27$	400	0.8500
3	$3\pi \approx 9.4248$	11.0880	$1200\pi \approx 3769.91$	600	0.5667
4	$4\pi \approx 12.5664$	14.7840	$1600\pi \approx 5026.55$	800	0.4250
5	$5\pi \approx 15.7080$	18.4800	$2000\pi \approx 6283.19$	1000	0.3400

3. Displacement modeshapes. From Problem B.3 (item 20), the displacement modeshapes are

$$\boxed{U_j(x) = \sin\left(\frac{j\pi x}{L}\right) = \sin\left(\frac{j\pi x}{0.85}\right), \qquad j = 1, \ldots, 5} \tag{B.4.3}$$

Figure 14 plots the displacement modeshapes for the first five modes across the tube length.

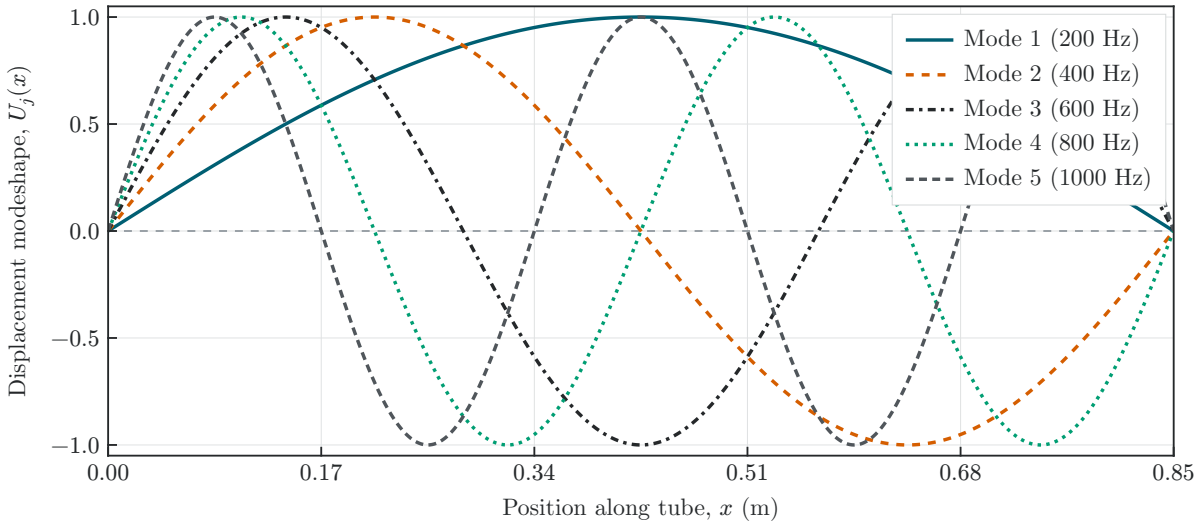


Figure 14: *Displacement modeshapes $U_j(x) = \sin(j\pi x/L)$ for modes $j = 1, \ldots, 5$ in an air tube of length $L = 0.85\,\mathrm{m}$ closed at both ends. Ticks along the horizontal axis mark the mode-5 nodal intervals ($x = jL/5$).*

4. Discussion of results.

- **Boundary condition verification:** Every displacement modeshape satisfies $U_j(0) = 0$ and $U_j(L) = 0$ identically. The rigid, impenetrable tube caps at $x = 0$ and $x = 0.85\,\mathrm{m}$ prevent fluid motion normal to the wall, establishing permanent displacement nodes at both ends.
- **Harmonic integer spectrum:** The fundamental resonant frequency is $f_1 = 200\,\mathrm{Hz}$. Because the eigenvalue spectrum satisfies $z_j = j\pi$, the higher resonance frequencies are exact integer multiples of the fundamental: $f_j = jf_1 = 200, 400, 600, 800, 1000\,\mathrm{Hz}$. Consequently, a closed-closed tube supports all integer harmonics (both even and odd).
- **Wavelength accommodation:** The tube length $L = 0.85\,\mathrm{m}$ accommodates an integral number of half-wavelengths for each mode: $L = j(\lambda_j/2)$. For the fundamental mode ($j = 1$), only one half of a wave spans the entire tube ($\lambda_1 = 1.70\,\mathrm{m} = 2L$). Each subsequent mode adds another half-wave within the same cavity length.
- **Nodes and antinodes:** Mode j contains exactly $j + 1$ displacement nodes (including the two ends) and j displacement antinodes where particle amplitude peaks at $|U_j| = 1$. The node-to-node spacing is precisely $\Delta x_{\mathrm{node}} = \lambda_j/2 = L/j$.

5. Graduate work: Normalized pressure modeshapes and comparison. From Problem B.3 (item 21), acoustic pressure disturbance is generated by fluid strain, $p(x,t) = -Ku_x(x,t) = -\rho c^2 u_x(x,t)$. For displacement amplitude $a_j > 0$ in meters, write $u_j(x,t) = a_j U_j(x) e^{-i\omega_j t}$. Differentiating $U_j(x) = \sin(\gamma_j x)$ gives $U_j'(x) = \gamma_j \cos(\gamma_j x)$, yielding the pressure amplitude

$$P_j(x) = -\rho c^2 a_j \gamma_j \cos\left(\frac{j\pi x}{L}\right) = -\rho c\, \omega_j a_j \cos\left(\frac{j\pi x}{L}\right).$$

Normalizing by the peak pressure amplitude $|P_j|_{\max} = \rho c\, \omega_j a_j$ yields the normalized pressure modeshape

$$\boxed{\bar{P}_j(x) = -\cos\left(\frac{j\pi x}{L}\right) = -\cos\left(\frac{j\pi x}{0.85}\right), \qquad j = 1, \ldots, 5} \tag{B.4.5}$$

Figure 15 plots the normalized pressure modeshapes for modes 1 through 5.

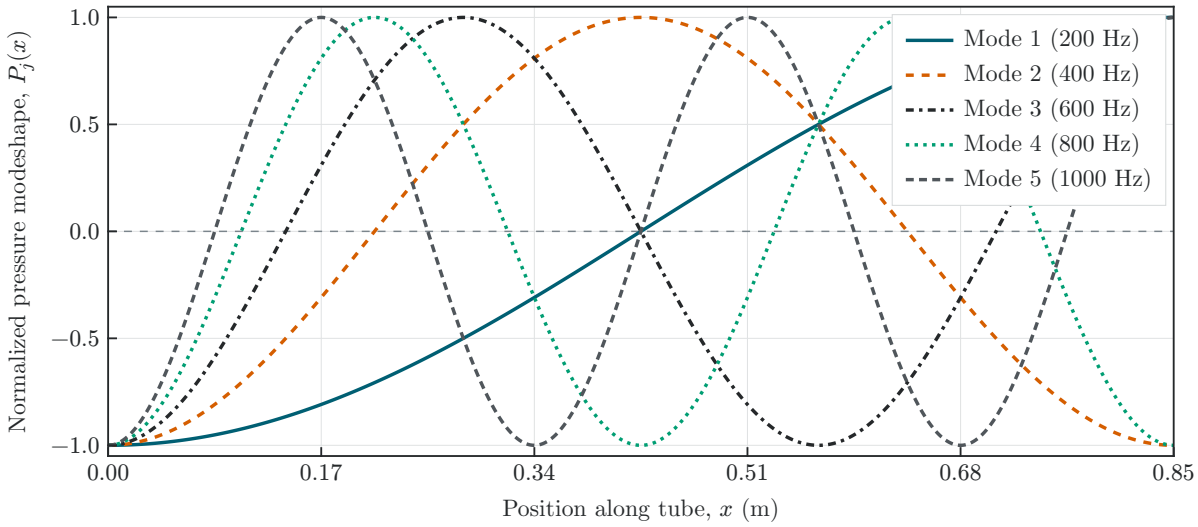


Figure 15: *Normalized acoustic pressure modeshapes $\bar{P}_j(x) = -\cos(j\pi x/L)$ for modes $j = 1, \ldots, 5$ in a closed-closed tube of length $L = 0.85\,\text{m}$.*

Comparison between displacement and pressure modeshapes:

(1) Spatial quadrature ($\pi/2$ phase shift): Spatial differentiation transforms $\sin(kx)$ into $k\cos(kx)$, introducing a spatial quarter-wavelength ($\pi/2$ radian) shift between displacement and pressure distributions.

(2) Boundary behavior: At the closed caps ($x = 0$ and $x = L$), displacement is strictly zero ($U_j = 0$, displacement nodes). The displacement gradient has maximum magnitude at these locations, so $p = -Ku_x$ gives maximum pressure magnitude. Consequently, the walls are always pressure antinodes ($|\bar{P}_j(0)| = 1$ and $|\bar{P}_j(L)| = 1$).

(3) Duality of nodes and antinodes: Wherever particle displacement achieves an antinode ($|U_j| = 1$), the spatial gradient is zero ($U_j' = 0$), meaning fluid volume change vanishes locally; thus, acoustic pressure disturbance is zero ($\bar{P}_j = 0$, pressure nodes). Conversely, at displacement nodes ($U_j = 0$), the spatial slope is steepest, creating maximal volumetric strain and therefore pressure antinodes ($|\bar{P}_j| = 1$).

(4) Nodal counts: For mode j, there are j internal pressure nodes (locations where the acoustic pressure perturbation is zero) and $j + 1$ pressure antinodes (including both rigid ends).

MATLAB code. *MATLAB script used to compute modal vibration characteristics and plot displacement and pressure modeshapes for Problem B.4.*

```
% B.4. Air vibration in a tube closed at both ends.
clear; clc; close all;

% 1. Input data
L = 0.85;           % tube length, m
c = 340;            % sound speed in air, m/s
rho = 1.2;          % air density, kg/m^3
N = 5;              % number of modes analyzed
j = (1:N)';

fprintf('=== Problem B.4: Air Vibration in Closed-Closed Tube ===\n');
fprintf('Input Data:\n');
fprintf('  Tube length L          = %.4f m\n', L);
fprintf('  Sound speed c          = %.1f m/s\n', c);
fprintf('  Air density rho        = %.2f kg/m^3\n', rho);
fprintf('  Bulk modulus K = rho*c^2 = %.1f kPa\n', (rho * c^2)/1e3);
fprintf('  Acoustic impedance Z   = %.1f Pa*s/m\n', rho * c);
fprintf('  Number of modes N      = %d\n\n', N);

% 2. Modal vibration characteristics
z = j * pi;                 % eigenvalues, rad
gamma = z / L;              % wavenumbers, rad/m
omega = gamma * c;          % natural angular frequencies, rad/s
f = omega / (2 * pi);       % natural frequencies, Hz
lambda = 2 * L ./ j;        % wavelengths, m

modal_table = table(j, z, gamma, omega, f, lambda, ...
    'VariableNames', {'Mode_j', 'Eigenvalue_z_j', 'Wavenumber_rad_m', ...
                      'Frequency_rad_s', 'Frequency_Hz', 'Wavelength_m'});
disp('Modal Vibration Characteristics:');
disp(modal_table);

% 3. Calculate and plot displacement modeshapes
Nx = 501;
x = linspace(0, L, Nx);
U = zeros(N, Nx);
P_norm = zeros(N, Nx);

for m = 1:N
    U(m, :) = sin(j(m) * pi * x / L);
    P_norm(m, :) = -cos(j(m) * pi * x / L);
end

% Palette colors matching course standard
C = [  0  95 115;   % deep teal
     213  94   0;   % orange
      37  40  42;   % graphite
       0 158 115;   % green
      80  86  91]/255; % slate

lineStyles = {'-', '--', ':', '-.', '--'};

figure('Name', 'Displacement Modeshapes', 'Color', 'w');
hold on;
for m = 1:N
```

```
    plot(x, U(m, :), 'Color', C(m, :), 'LineStyle', lineStyles{m}, ...
         'LineWidth', 1.4, 'DisplayName', sprintf('Mode %d (%d Hz)', m, round(f(m))));
end
yline(0, ':k', 'Alpha', 0.5);
xlabel('Position along tube, x (m)');
ylabel('Displacement modeshape, U_j(x)');
title(sprintf('Air Displacement Modeshapes (L = %.2f m, Closed-Closed)', L));
xlim([0, L]); ylim([-1.1, 1.1]);
xticks(linspace(0, L, 6));
legend('Location', 'northeast');
grid on; box on;

% 5. Graduate work: Calculate and plot normalized pressure modeshapes
figure('Name', 'Normalized Pressure Modeshapes', 'Color', 'w');
hold on;
for m = 1:N
    plot(x, P_norm(m, :), 'Color', C(m, :), 'LineStyle', lineStyles{m}, ...
         'LineWidth', 1.4, 'DisplayName', sprintf('Mode %d (%d Hz)', m, round(f(m))));
end
yline(0, ':k', 'Alpha', 0.5);
xlabel('Position along tube, x (m)');
ylabel('Normalized pressure modeshape, P_j(x)');
title(sprintf('Normalized Acoustic Pressure Modeshapes (L = %.2f m, Closed-Closed)', L));
xlim([0, L]); ylim([-1.1, 1.1]);
xticks(linspace(0, L, 6));
legend('Location', 'northeast');
grid on; box on;
```

B.5 Calculate the Slide Whistle

Consider the slide whistle shown in Figure 16. The fully retracted whistle is shown in Figure 16(a), whereas the fully extended whistle is shown in Figure 16(b). The length of the residual cavity of the fully retracted whistle tube can be approximated to be $L_0 = 6\,\text{mm}$ (Figure 16(a)). The thickness of the piston inside the whistle can be approximated to be $L_p = 6\,\text{mm}$. The thickness of the end walls can be approximated to be $L_w = 6\,\text{mm}$. Assume sound speed in air to be $c = 340\,\text{m/s}$. The purpose of this problem is to calculate the fundamental frequency for various positions of the whistle slider



(a)



(b)

Figure 16: *Slide whistle. (a) Fully retracted, showing the piston end. (b) Fully extended. The ruler's major markings are in centimeters.*

Recall the theoretical results of Problem B.3.

Do the following.

(1) Display input data.

(2) Use Figure 16 to estimate the fully extended whistle length L_e.

(3) Use Figure 16 to estimate the fully contracted whistle length L_c.

(4) Use L_e, L_c to calculate the extension travel dL_I corresponding to fully extended slide whistle.

(5) Use dL_I and L_0 to calculate the cavity length L_I of a fully extended slide whistle.

(6) Estimate the fully extended frequency f_I in Hz and find the nearest musical note expressed as a letter name and octave number (e.g., G3 signifies note G in the third octave).

(7) Use L_e, L_c to calculate the extension travel dL_{II} corresponding to 40% extended slide whistle.

(8) Use dL_{II} and L_0 to calculate the cavity length L_{II} of a 40%-extended slide whistle.

(9) Calculate the 40%-extended frequency f_{II} in Hz and find the nearest musical note expressed as a letter name and an octave number.

(10) Comment on your work.

Graduate Work

Estimate the piston position required to obtain the musical note B7.

Do the following.

(11) Display the frequency f_{III} of the note B7.

(12) Calculate the cavity length L_{III} needed to generate frequency f_{III} of the note B7.

(13) Calculate the extension travel dL_{III} needed to generate frequency f_{III} of the note B7.

(14) Estimate what percentage of the piston must be extended to generate the note B7.

(15) Comment on your work.

Solution to B.5

1. Input data. The specified dimensions and sound speed are

$$\boxed{L_0 = 6\,\text{mm} = 0.006\,\text{m}, \qquad L_p = 6\,\text{mm}, \qquad L_w = 6\,\text{mm}, \qquad c = 340\,\text{m/s}.}$$

The ruler markings in Figure 16 are used as the reference scale. I measured the overall whistle from the left mouthpiece tip to the right piston-handle tip. This gives the estimates

$$\boxed{L_e \approx 0.210\,\text{m}, \qquad L_c \approx 0.135\,\text{m}.}$$

The photograph provides an overall length estimate rather than a direct measurement of the internal air-column endpoints. The fixed piston and wall thicknesses therefore remain input dimensions, while their contributions cancel when the change in overall position is calculated.

2. Fully extended whistle length. Reading the upper ruler in Figure 16(b) from the left mouthpiece tip to the right handle tip gives

$$\boxed{L_e \approx 0.210\,\text{m}.}$$

3. Fully contracted whistle length. Reading the upper ruler in Figure 16(a) between the same two physical reference points gives

$$\boxed{L_c \approx 0.135\,\text{m}.}$$

4. Fully extended travel. The fixed wall and piston thicknesses are present in both overall-length measurements. Thus, the slider travel is

$$\boxed{dL_{\text{I}} = L_e - L_c = 0.210 - 0.135 = 0.075\,\text{m}.}$$

5. Fully extended cavity length. The problem directs us to recall the closed-closed model from Problem B.3. Using the residual cavity length as the fixed offset gives

$$\boxed{L_{\text{I}} = L_0 + dL_{\text{I}} = 0.006 + 0.075 = 0.081\,\text{m}.}$$

6. Fully extended frequency. For the assigned closed-closed model, the fundamental frequency is $f = c/(2L)$. Therefore,

$$f_{\text{I}} = \frac{340}{2(0.081)} = 2098.77\,\text{Hz} \approx 2.10\,\text{kHz}.$$

The nearest equal-tempered note is

$$\boxed{f_{\text{I}} \approx 2.10\,\text{kHz}, \qquad \text{nearest note C7 } (2093.00\,\text{Hz}).}$$

7. Forty-percent extension travel. The travel at 40% extension is

$$\boxed{dL_{\mathrm{II}} = 0.40 dL_{\mathrm{I}} = 0.40(0.075) = 0.030\,\mathrm{m}.}$$

8. Forty-percent extension cavity length. Adding the residual cavity gives

$$\boxed{L_{\mathrm{II}} = L_0 + dL_{\mathrm{II}} = 0.006 + 0.030 = 0.036\,\mathrm{m}.}$$

9. Forty-percent extension frequency. The closed-closed fundamental is

$$f_{\mathrm{II}} = \frac{340}{2(0.036)} = 4722.22\,\mathrm{Hz}.$$

The nearest equal-tempered note is

$$\boxed{f_{\mathrm{II}} \approx 4.72\,\mathrm{kHz}, \qquad \text{nearest note D8 } (4698.64\,\mathrm{Hz}).}$$

10. Comment. The ruler estimates give a travel of approximately 75 mm, and the model predicts that increasing the cavity length lowers the fundamental frequency. The two frequencies follow this inverse relationship. A physical slide whistle has an open fipple and a closed piston, so the course notes' closed-open model (pp. 114–117) gives $f = c/(4L)$ for that configuration. The corresponding estimates would be 1049.38 Hz (approximately C6) at full extension and 2361.11 Hz (approximately D7) at 40% extension. The closed-closed results above follow the assignment instruction to recall Problem B.3; the open-fipple behavior is the relevant physical limitation of that idealization [5].

11. Frequency of B7. Relative to A4, B7 is 38 semitones higher. Equal temperament gives

$$\boxed{f_{\mathrm{III}} = 440\,2^{38/12} = 3951.07\,\mathrm{Hz} \approx 3.95\,\mathrm{kHz}.}$$

12. Cavity length for B7. Applying the assigned closed-closed fundamental relation gives

$$\boxed{L_{\mathrm{III}} = \frac{c}{2f_{\mathrm{III}}} = \frac{340}{2(3951.07)} \approx 0.0430\,\mathrm{m}.}$$

13. Extension travel for B7. The required travel beyond the residual cavity is

$$\boxed{dL_{\mathrm{III}} = L_{\mathrm{III}} - L_0 = 0.04303 - 0.006 \approx 0.0370\,\mathrm{m}.}$$

14. Required piston extension. Relative to the full measured travel, the required percentage is

$$\boxed{100\frac{dL_{\mathrm{III}}}{dL_{\mathrm{I}}} = 100\frac{0.03703}{0.075} = 49.4\%.}$$

15. Comment. The B7 calculation places the slider at approximately half of the available measured travel under the assigned closed-closed model. The result is within the photographed travel range. Using the physical closed-open fipple model would require $L_{\mathrm{III}} = 0.02151\,\mathrm{m}$, or approximately 20.7% of the measured travel. Thus, the boundary-condition model determines the numerical prediction.

MATLAB code. *MATLAB calculations for B.5.*

```
% B.5. Slide whistle frequency estimates.
clear; clc;

L0 = 0.006;                % residual cavity length, m
Lp = 0.006;                % piston thickness, m
Lw = 0.006;                % end-wall thickness, m
c = 340;                   % sound speed, m/s
Le = 0.210;                % measured extended overall length, m
Lc = 0.135;                % measured contracted overall length, m

dLI = Le - Lc;             % full slider travel, m
LI = L0 + dLI;             % fully extended cavity, m
fI = c / (2 * LI);         % closed-closed fundamental, Hz

dLII = 0.40 * dLI;         % 40 percent travel, m
LII = L0 + dLII;           % 40 percent cavity, m
fII = c / (2 * LII);       % 40 percent fundamental, Hz

fIII = 440 * 2^(38/12);    % B7, Hz
LIII = c / (2 * fIII);     % cavity for B7, m
dLIII = LIII - L0;         % travel for B7, m
pct = 100 * dLIII / dLI;   % percent of full travel

n = round(69 + 12*log2([fI fII]/440)); % nearest MIDI note numbers
s = ["C" "C#" "D" "D#" "E" "F" "F#" "G" "G#" "A" "A#" "B"];
notes = s(mod(n,12)+1) + string(floor(n/12)-1);

fprintf('B.5 Slide Whistle\n');
fprintf('Inputs: L0 = %.3f m, Lp = %.3f m, Lw = %.3f m, c = %.1f m/s\n', ...
    L0, Lp, Lw, c);
fprintf('Measured lengths: Le = %.3f m, Lc = %.3f m\n', Le, Lc);
fprintf('Full travel = %.3f m, cavity = %.3f m, frequency = %.2f Hz (%s)\n', ...
    dLI, LI, fI, notes(1));
fprintf('40%% travel = %.3f m, cavity = %.3f m, frequency = %.2f Hz (%s)\n', ...
    dLII, LII, fII, notes(2));
fprintf('B7 frequency = %.2f Hz, cavity = %.5f m, travel = %.5f m (%.1f%%)\n', ...
    fIII, LIII, dLIII, pct);

% Physical open-fipple, closed-piston comparison.
fo = [fI fII]/2;
lo = c/(4*fIII);
po = 100*(lo-L0)/dLI;
fprintf('Closed-open comparison: %.2f Hz, %.2f Hz; B7 at %.1f%% travel\n', ...
    fo(1), fo(2), po);
```

B.6 Calculate Sound Intensity and the dB Scale

Take sound speed $c = 340\,\mathrm{m/s}$ and air density $\rho = 1.2\,\mathrm{kg/m^3}$.

Do the following.

(1) Display input data.

(2) Calculate air's acoustic impedance Z.

(3) Calculate the sound intensity I corresponding to a sound pressure $p = 55\,\mathrm{mPa}$.

(4) Give the values of the reference sound intensity I_0 and reference sound pressure p_0.

(5) Convert the sound pressure and sound intensity from item (3) to decibels.

(6) Verify that the sound pressure and sound intensity calculated at item (5) have the same dB value and explain why it is so.

(7) Calculate in physical units the sound intensity, sound pressure, and sound power from a loudspeaker with diameter $D = 1.55\,\mathrm{m}$ working at $78\,\mathrm{dB}$.

(8) Calculate the sound pressure p_{horn} created by a $118\,\mathrm{dB}$ car horn.

(9) Give the values of the pain sound intensity I_{pain} and pain sound pressure p_{pain} and comment on the results of items (7) and (8) relative to pain values.

(10) Graduate work. Explain why different decibel-conversion formulas are required for sound intensity I and sound pressure p.

3 Extra Credit Challenge Problems

Challenge problems are optional.

C.1 Analyze Air Vibration in a Closed-Open Tube

(1) Sketch a tube closed at one end and open at the other end indicating all the information needed for the analysis.

(2) Write the boundary conditions at the two ends of the tube in terms of displacement $u(x,t)$ and pressure disturbance $p(x,t)$.

(3) Recall from HW01 the harmonic solution of the wave equation consisting of the multiplication of two terms:

- **(a)** First term: a function $U(x)$ consisting of trigonometric functions depending on wavenumber γ, space variable x, and constants C_1, C_2.
- **(b)** Second term: a complex exponential depending on frequency ω and time variable t.

(4) Recall from HW01 the expression of pressure $p(x,t)$ in terms of displacement $u(x,t)$ and bulk modulus K.

(5) Substitute (3) into (4) to obtain the pressure $p(x,t)$ in terms of constants C_1, C_2.

(6) Substitute (3) and (5) into (2) and write the boundary conditions at the two ends of the tube in terms of unknown constants C_1, C_2.

(7) Solve (6) for C_2 and obtain an equation for C_1.

(8) Discuss the trivial solution.

(9) Choose to search for the nontrivial solution and define the equation that needs to be solved to find a nontrivial solution.

(10) Introduce the notation $z = \gamma L$ into (9) and write the eigenvalue equation in terms of z.

(11) Solve the eigenvalue equation and write the general expression for the eigenvalues z_j, where $j = 1, 2, 3, \ldots$.

(12) Use (10) to write the general expression for the wavenumber γ_j in terms of j and length L.

(13) Recall from HW01 the definition of wavenumber γ in terms of frequency ω and wavespeed c.

(14) Solve (13) for frequency ω in terms of wavenumber γ and wavespeed c.

(15) Use (12) and (14) to write the general expression of the natural frequency ω_j in terms of j, wavespeed c, and tube length L.

(16) Use (15) to write the general expression of the Hz frequency f_j in terms of j, wavespeed c, and tube length L.

(17) Explain the meaning of fundamental frequency and overtones.

(18) Explain the meaning of fundamental harmonic and higher harmonics.

(19) Recall from HW01 the relationship between wavelength λ and wavenumber γ.

(20) Substitute (12) into (19) to write the general expression for the wavelength λ_j in terms of j and tube length L.

(21) Use (20) to explain which wavelength multiples could accommodate vibratory standing waves in a tube of length L closed at one end and open at the other end.

(22) Recall $U(x)$ from (3)(a) and use (7) to eliminate C_2.

(23) Set $C_1 = 1$ in item (22) and write the general expression for the modeshape $U_j(x)$ in terms of the wavenumber γ_j.

(24) Plot the normalized displacement modeshapes for modes 1 through 5 of air vibration in a tube of length L, and discuss your results.

(25) Graduate work. Calculate and plot the normalized pressure modeshapes for modes 1 through 5 of air vibration in a tube of length L, and discuss your results.

C.2 Analyze Air Vibration in an Open-Open Tube

(1) Sketch a tube open at both ends indicating the information needed for analysis.

(2) Write the boundary conditions at the two ends of the tube.

(3) Repeat all steps of Problem C.1 paying particular attention to the values taken by C_1 and C_2.

C.3 Calculate Air Vibration in a Tube with Various Boundary Conditions

Consider a tube of length $L = 0.85\,\text{m}$. Take sound speed $c = 340\,\text{m/s}$ and air density $\rho = 1.2\,\text{kg/m}^3$. The purpose is to calculate the modal vibration characteristics and modeshapes for $j = 1, \ldots, N$ with $N = 5$. The following boundary conditions are considered.

Case	$x = 0$	$x = L$
(i)	Closed	Open
(ii)	Open	Open

(1) Display the input data.

For each boundary-condition case, complete items (2)–(5).

(2) Calculate and display the modal vibration characteristics:

- **(a)** Eigenvalues z_j.
- **(b)** Wavenumbers γ_j.
- **(c)** Rad/s frequencies ω_j.
- **(d)** Hz frequencies f_j.

(3) Calculate and plot the displacement modeshapes.

(4) Compare your results with those of Problem B.4.

(5) Graduate work. Calculate and plot the normalized pressure modeshapes, and compare the result with item (3).

References

[1] National Aeronautics and Space Administration. Our high-energy universe: 15 years with the Fermi Gamma-ray Space Telescope. Technical Report SP-2024-4-254-GSFC, NASA's Goddard Space Flight Center, Greenbelt, Maryland, 2024. https://science.nasa.gov/wp-content/uploads/2024/04/our-high-energy-universe.pdf.

[2] Tamara Smyth. Properties of sound waves. Music 270a: Fundamentals of Audio, Acoustics and Sound, Department of Music, University of California, San Diego, 2019. Accessed October 16, 2026. https://musicweb.ucsd.edu/~trsmyth/funAudio/Properties_Sound_Waves.html.

[3] The Nobel Prize. Waves or particles? Einstein explains the photoelectric effect. Nobel Prize Educational. Accessed October 16, 2026. https://educationalgames.nobelprize.org/educational/physics/quantised_world/waves-particles-2.html.

[4] Eite Tiesinga, Peter J. Mohr, David B. Newell, and Barry N. Taylor. CODATA recommended values of the fundamental physical constants: 2018. *Journal of Physical and Chemical Reference Data*, 50(3):033105, 2021. https://physics.nist.gov/cuu/pdf/JPCRD2018CODATA.pdf.

[5] University of California, Santa Barbara. 44.27—wind instruments. Physics Lecture Demonstrations. Online demonstration. Accessed October 18, 2026.

[6] Joe Wolfe. Brass instrument (lip reed) acoustics: An introduction. Music Acoustics, University of New South Wales. Accessed October 17, 2026. https://www.phys.unsw.edu.au/jw/brassacoustics.html.

[7] Joe Wolfe. How do woodwind instruments work? Music Acoustics, University of New South Wales. Accessed October 17, 2026. https://newt.phys.unsw.edu.au/jw/woodwind.html.